

SPARSE OBSERVATION ALGEBRAS: COLLISION GEOMETRY AND SHARED MEMORY

QIAN QI

ABSTRACT. We determine the persistent memory of positive sparse monomial experiments through arbitrary additive exponent collisions, and prove a composition law when independent experiments share a single finite-state memory. At a scalar checkpoint, the optimal squared-prediction regret is comparable to the maximum of partial Vandermonde-volume powers, truncated at the attainable past dimension. The comparison is uniform on compact subsets of the one-step exponent chamber, including exact collisions, for fixed full-support priors and finite horizons. Under independent composition, the future query space is a tensor product, but the attained information dimensions and resolved bit requirements add at each checkpoint. This holds against arbitrary joint encoders, not only product codes. One index-only transducer attains the maximum checkpoint law through all stages. The proof combines complete Newton–Hermite flags, a global bounded-format cover retaining each block’s past cap, a product minorization with acquisition evidence, and nonsingular raw moment updates. The resulting bit law quantifies the effect of observation order: serial schedules attain the largest individual memory requirement, whereas overlapping schedules can attain their sum. Collision-tree formulas and the exact, affine, control and common-risk consequences are proved under their respective operational hypotheses.

1. INTRODUCTION

A finite experiment creates information at one rate and uses it at another. The information that must persist is determined by the tests that remain available, not by the full accumulated likelihood alone. For positive sparse monomial experiments we compute this persistent information, both exactly and at prescribed accuracy. We then determine how the answer composes when several independent experiments share one memory and their observations are interleaved.

There are two different sources of variation. Additive coincidences among future exponents change which directions can be tested. Near a coincidence, a direction can be exactly observable but have arbitrarily small statistical amplitude. Independently, the order in which several experiments are observed changes how many of their attained directions must be retained at the same time. The first phenomenon requires a collision-uniform resolution law. The second requires a joint coding converse, not merely the construction of separate block quantizers.

1.1. The scalar resolution law. Let

$$A = \{0 = a_0 < a_1 < \cdots < a_{r-1} = D\}, \quad r \geq 2, \quad W_A = \text{span}\{t^a : a \in A\}.$$

Write mA for sums of exactly m elements of A , including zero padding, and $h_A(m) = |mA|$, with $0A = \{0\}$. A positive finite detector has cells spanning W_A . Its commands are independent lookup entries in a fixed cube $[\eta, 1 - \eta]^J$, $0 < \eta < 1/2$; the accepted cell label or the failure symbol is reported. The full experiment is defined in Section 2.

Date: September 6, 2026.

2020 Mathematics Subject Classification. 62B05, 62C10, 93E11, 13D40.

Key words and phrases. attainable information state, sparse observation algebra, exponent collision, finite-state filtering, joint quantization, shared memory.

For the uniform assertion the hidden scalar lies in $[0, 1]$ with one fixed full-support prior μ . The known nonzero exponents range over a compact set K inside $0 < a_1 < \dots < a_{r-1}$. A fixed coefficient matrix of column rank r realizes the cells throughout K ; their sum is one and their common lower bound is positive. Fix a horizon N and $H \geq \max\{1, N \max_K a_{r-1}\}$. At a checkpoint with n past and $m = N - n$ remaining trials, the positive formal homogeneous labels $\alpha \in \mathbb{N}_0^r$, $|\alpha| = m$, $\alpha \neq me_0$, give nodes $x_\alpha = H^{-1} \sum_i \alpha_i a_i$. Their number is q_m . They are retained as formal labels when their values coincide. Put

$$\mathcal{V}_{m,\ell}(a) = \max_{|J|=\ell} \prod_{\{\alpha,\beta\} \subset J} |x_\alpha - x_\beta|, \quad \mathcal{V}_{m,1} = 1, \quad p_{n,m} = \min\{n(r-1), q_m\},$$

where J ranges over positive formal labels. A volume is zero if fewer than ℓ distinct sites remain. Define

$$\Psi_{n,m}(M, a) = \max_{1 \leq \ell \leq p_{n,m}} (\mathcal{V}_{m,\ell}(a)/M)^{2/\ell}, \quad \Xi_N(M, a) = \max_{1 \leq n < N} \Psi_{n,N-n}(M, a),$$

with zero checkpoint profile when $n = 0$ or $m = 0$.

Theorem 1.1 (Intrinsic attainable resolution and causal memory). *Under these hypotheses there are constants $0 < c \leq C < \infty$, depending only on the fixed experiment, K , N and μ , with the following properties for every known calibration $a \in K$ and every integer $M \geq 1$.*

At each checkpoint $n + m = N$, the optimal M -message squared-prediction regret is between $c\Psi_{n,m}(M, a)$ and $C\Psi_{n,m}(M, a)$, both unconditionally under independent uniform acquisition commands and in the minimax sense over admitted histories. The query is sampled independently after the index is formed, from the ordered products of one fixed attainable failure-factor basis; all its m trials are counted.

There exists one deterministic stage-compatible transducer, allowed to depend on a and M , storing at most M persistent labels after every report, whose regret at every checkpoint and on every history is at most $C\Xi_N(M, a)$. Conversely every such transducer has maximum unconditional checkpoint regret at least $c\Xi_N(M, a)$ under the same common exploration law. Independent coding randomization does not remove the lower bound. The clock and fixed real-valued program are read-only as in Definition 3.1.

The constants are uniform across all additive collision strata of K . No calibration-blind codebook or uniformity over all full-support priors is part of the assertion.

The proof has three parts. At a common interior product of binomials, the tangent pairs at full rank with every initial Newton–Hermite flag, including at repeated nodes. Actual exploration histories therefore minorize a uniformly sized resolved cube, with their acquisition evidence retained. A bounded-format covering estimate controls the entire reachable image, with the products of scales truncated at the past capacity. Finally, raw moment updates have no inverse exponent gaps, so one persistent index realizes these covers through all stages. These parts are proved in that order in Sections 4–7.

The determinant profile has classical finite spectral content. If W_m is the square ordinary Vandermonde matrix on the positive formal nodes, then $\mathcal{V}_{m,\ell} \asymp \|\bigwedge^\ell W_m\|$, uniformly even at coincidences; Proposition 5.3 records the precise comparison. It does not prove attainable posterior geometry or compatibility with causal compression. The arbitrary-prefix Hermite fact is likewise isolated as Lemma 5.1 before it is used in that geometry.

1.2. Composition with one shared memory. Take B independent scalar experiments of this class, with possibly different detectors, exponent sets, priors and horizons N_b . Their joint hidden parameter is a vector with the product prior. A deterministic schedule ω specifies which block is observed on each of $T = \sum_b N_b$ trials. At time s its count vector is $(n_b(s))$, with

$m_b(s) = N_b - n_b(s)$. A future query chooses a product probe in every block and predicts the all-failure event of the resulting tensor query. All $T - s$ remaining trials are executed. There is one shared memory index, not an uncharged tuple of separate block memories.

At a block checkpoint put $V_{b,0} = 1$ and $V_{b,\ell} = \mathcal{V}_{b,m_b,\ell}(a^{(b)})$ for $1 \leq \ell \leq p_b = \min\{n_b(r_b - 1), q_{b,m_b}\}$. When $n_b = 0$ or $m_b = 0$ put $p_b = 0$. Define the real bit profile

$$b_b(n_b, \varepsilon, a^{(b)}) = \max_{0 \leq \ell \leq p_b} \left\{ \frac{\ell}{2} \log_2 \frac{1}{\varepsilon} + \log_2 V_{b,\ell} \right\}.$$

The zero-index term is zero and zero positive-index volumes are omitted.

Theorem 1.2 (Shared-memory information law). *For these fixed block experiments, the least integer persistent bit budget for maximum-checkpoint squared-prediction regret at most ε is*

$$(1) \quad B_{\omega}^*(\varepsilon, a) = \max_s \sum_b b_b(n_b(s), \varepsilon, a^{(b)}) + O(1), \quad 0 < \varepsilon \leq 1.$$

This holds both for maximum-history regret and for unconditional regret under one common independent uniform exploration law. The bounded remainder is uniform in accuracy, calibration and schedule. The lower bound permits arbitrary joint randomized encoders; the upper bound is realized by one stage-compatible index-only transducer.

Writing $L_b = \max_n b_b(n, \varepsilon, a^{(b)})$, the smallest possible budget over schedules is $\max_b L_b + O(1)$, attained by every serial schedule. The largest is $\sum_b L_b + O(1)$, attained in order by a schedule that brings all blocks to their respective maximizing counts. A fixed serial schedule is optimal in this comparison sense simultaneously for every calibration and accuracy.

The all-budget regret form and complete proof are Theorem 8.3; its logarithmic inversion and scheduling consequences are Theorems 9.2 and 9.3. The joint attainable image is a product embedded by a multilinear map, not a full tensor-dimensional body. Its metric is recovered from the full tensor query menu by a bounded linear map defined even on arbitrary decoded centers. A product of blockwise dimension-truncated covers yields the upper bound, while a product of attained subprobability rectangles yields a converse against unrestricted joint coding. The raw updates then prove causal compatibility. Thus (1) is not obtained by defining the optimum to be a product code.

This gives an observation-order effect with a sharp change of exponent. For B independent two-exponent blocks, each with horizon $2h$, a serial schedule has streaming regret $\asymp M^{-2/h}$; a schedule bringing all blocks to their middle checkpoint has regret $\asymp M^{-2/(Bh)}$. The number of raw trials is $2Bh$ in both cases and the query protocol is the same full tensor construction (Corollary 9.4). The saving uses the specified future-only task: it does not erase an externally requested past payoff.

1.3. Exact sufficiency and the proof structure. The exact scalar law also holds on a fixed interval $I = [l, u]$, $0 \leq l < u < \infty$, with a full-support prior. Its role is to identify the attained tangent before any finite-error comparison is made.

Theorem 1.3 (Sparse finite-horizon information law). *Suppose the positive finite detector spans W_A and every lookup entry ranges over $[\eta, 1 - \eta]$, where $0 < \eta < 1/2$. For $n, m \geq 0$ put*

$$(2) \quad d_A(n, m) = \min\{n(r - 1), h_A(m) - 1\}.$$

Two attainable posteriors after n reports have the same law under every common future-only continuation of length m if and only if their moments at the exponents in mA agree. The prediction map on all-failure command histories has generic rank $d_A(n, m)$, and it has a smooth attainable patch of this dimension with a continuous section to histories. The smallest Euclidean

dimension of a sufficient encoding continuous on each known-report history stratum is exactly $d_A(n, m)$.

For a total budget N , these dimensions can be attained simultaneously by a causal exact-real state. Its maximal size is

$$(3) \quad D_A(N) = \max_{0 \leq n \leq N} d_A(n, N - n).$$

Only prior moments with exponents in NA are needed for its updates.

For independent blocks the exact dimensions add at a checkpoint and their sum is then maximized over the schedule (Theorem 8.2). Continuous encodings here are encodings of known-command histories on each report stratum. They need not identify different factorizations of the same posterior, and no global chart of a singular quotient is assumed.

The principal proof proceeds from the observation experiment and memory model to the product tangent, complete Hermite flags, the global covering estimate, the collision-uniform law, and finally the shared-state law. The observation-algebra calculations and the full affine, five-trial, seven-trial, control, decision and mechanical arguments are given in Appendices A–H. Their distinct operational hypotheses are not replaced by the product experiment. In particular, the binary ticket identity transfers prediction loss to one specified common payoff; it is not a universal optimal-control lower bound. The separate control upper bound retains its weaker exponent.

1.4. Relation to the classical ingredients. Predictive state representations, approximate recursively updated information states and squared-error quantization are established subjects [6, 8, 5, 4]. The mixed and confluent determinant arguments use classical total positivity [7]. The sumset interpretation of a homogeneous observation algebra is also classical [2, 3]. The Newton–Hermite prefix principle is explicitly stated in [12, Proposition 7, p. 48]; finite Leja orderings and their linear-algebraic implementations are classical [20, 21]. The real covering input is separated in Lemma 6.1, using [16, 17, 18].

Clustered Fourier Vandermonde spectra are studied in [15, Theorems 2.2–2.3]; the matrix comparison used here concerns an ordinary finite real Vandermonde matrix and is proved directly. Neither comparison supplies the attained history distribution or the persistent-index converse. Functional product quantization [22, Sections 1.1–1.2 and 5.5] optimizes output error when input components are separately quantized. In the present problem the histories are generated sequentially, the future menu spans a tensor observation algebra, and the converse permits an arbitrary joint M -message code. Its uniform constants include exponent coincidences, while its one-index upper bound accumulates all preceding compression errors. The elementary multiplication of covering estimates is used as such, rather than presented as a new general quantization principle.

The finite spectral comparison and the convex allocation argument in Appendix H were made explicit in the source-pinned technical note [23]; they are not reassigned as independent contributions of the composition theorem. The specific attainable pairing, the global image bound, the joint coding converse and their causal compatibility are the mathematical assertions proved here. The full earlier mechanical and likelihood derivations remain separate sources [9, 13, 11]; no long-time conclusion is imported from their historical labels into a fixed-horizon proof.

All uniform constants in the resolved theorems are for fixed priors, detectors, command cubes and finite horizons. Each known calibration and budget may have its own read-only exact-real program. The product theorem requires a product prior. Neither prior-uniform conditioning,

effective codebook synthesis, calibration-blind filtering nor arbitrary hidden- parameter terminal losses are implicit in these statements.

2. CALIBRATED EXPERIMENTS AND THEIR OPERATIONAL QUOTIENT

All spaces carry their Borel sigma-algebras. Deterministic continuation, transition and decision rules are Borel measurable; randomized rules are Borel probability kernels. Every external random seed is independent of the hidden parameter and the acquired history unless explicitly charged as part of that history.

Let $k_1, \dots, k_K \in W_A$ be strictly positive on I , satisfy $\sum_j k_j = 1$, and span W_A . On each trial a fresh raw symbol has these probabilities conditional on the same fixed parameter t . The raw trials are conditionally independent. A command $g \in \mathcal{G} = [\eta, 1 - \eta]^K$ accepts raw symbol j with probability g_j , using an independent prepared uniform variable. A rejected symbol is reported as $\dagger$. Thus the report alphabet is $\mathcal{X} = \{1, \dots, K, \dagger\}$ and its likelihood factors are

$$(4) \quad f_{g,j} = g_j k_j, \quad f_{g,\dagger} = F_g = \sum_j (1 - g_j) k_j.$$

A single command is applied at every possible parameter value. Every trial is counted, including every rejected report. There is a constant $\kappa > 0$ such that $\kappa \leq f_{g,x}(t) \leq 1$ for all g, x, t . Indeed one may take $\kappa = \eta \min_{j,t} k_j(t)$.

Lemma 2.1 (Attainable interior and realizability). *The failure family is precisely the image of the rejection cube under $u \mapsto \sum_j u_j k_j$. It contains a neighborhood of the constant $1/2$ in W_A . Conversely, every finite exponent set A as above has a positive normalized r -cell realization with this property.*

Proof. The displayed map is a surjective linear map onto W_A . Choose a linear right inverse $L : W_A \rightarrow \mathbb{R}^K$. For every v satisfying $\|Lv\|_\infty < 1/2 - \eta$, rejection vector $u = \mathbf{1}/2 + Lv$ realizes $1/2 + v$ inside the cube. This proves both the interior statement and that the same physical command implements it. For existence, choose $\delta > 0$ with $\delta \sum_{a \in A \setminus \{0\}} u^a < 1/r$ and set

$$k_a(t) = 1/r + \delta t^a \quad (a \neq 0), \quad k_0(t) = 1/r - \delta \sum_{a \neq 0} t^a.$$

They are positive and sum to one. Their span contains one and then each $t^a = (k_a - 1/r)/\delta$, so has dimension r . These are calibrated categorical experiments. No mechanical origin is asserted for this general realization; the separate Lorentz realization is derived in Section G. $\square$

A history $h = (g_1, x_1, \dots, g_n, x_n)$ has posterior

$$(5) \quad P_h(t) \mu(dt), \quad P_h = \frac{\prod_{i=1}^n f_{g_i, x_i}}{\mu(\prod_{i=1}^n f_{g_i, x_i})}.$$

All commands are known inputs when received. They cease to be available once discarded by a memory-limited filter. A continuous history encoding means continuity in $(g_1, \dots, g_n)$ separately for each fixed report word. It need not be injective on histories and need not identify all histories with the same posterior.

Definition 2.2 (Common future-only continuation). At a checkpoint a continuation has the same fixed task data in the two experiments. Its commands are functions only of the reports and commands *since that checkpoint*, and any fresh independent random seed. Its first command is the same in both experiments. It may not read the original prefix, a prefix-dependent seed, or any other uncharged historical tape. Two posteriors are m -equivalent when all such length- m continuations have the same future report law.

A separate payoff automaton referring to past observations is task memory, not something removed by posterior equivalence. Losses that inspect t directly are also not included without adding their test functions. This quantifier matters: two different constant failure commands leave the same prior, but a rule that reads and repeats the original command does not meet Definition 2.2.

Proposition 2.3 (Exact future test space). *The linear span of length- m future likelihood tests is*

$$(6) \quad W_A^m = \text{span}\{t^s : s \in mA\}.$$

Two posteriors are m -equivalent if and only if their moments at the exponents in $mA \setminus \{0\}$ agree. A finite list of actually executable, nonadaptive all-failure probes spans the same space.

Proof. Choose finitely many attainable failures spanning W_A , using Lemma 2.1. All their m -fold products are executable all-failure event probabilities. Multilinearity makes their span the span of all products of m members of W_A , which is exactly (6). Independence is conditional on t ; one does not replace the shared parameter by fresh parameters between trials.

For any fixed future word a deterministic continuation chooses a known sequence of commands along that word. Its likelihood is a product of m members of W_A , even when the chosen sequence differs on other branches. Equality of the stated moments therefore implies equality of all word probabilities. There are finitely many report words. Integrating an independent continuation seed proves the randomized case. Conversely, equivalence includes all the spanning all-failure probes; their predictions determine all the moments in (6). The constant moment is one. Only the selected probe is executed and all its m trials are consumed, regardless of earlier nonfailure reports. $\square$

A fixed positive nonconstant denominator or an arbitrary nonmonomial factor space is not silently substituted in (6). Such models have their own product and test pairings. The theorem below addresses every sparse monomial space W_A , not every finite-dimensional space of continuous functions.

3. PERSISTENT MEMORY AND THE PREDICTION TASK

Fix $N \geq 2$ and abbreviate $d_n = d_A(n, N - n)$ and $D_* = D_A(N) > 0$. The exact state from Theorem 4.4 will be denoted s_n . A clock n , calibration, prior moments and a precomputed finite program are read-only data. The charged resource is the persistent state between successive input symbols.

Definition 3.1 (Finite-state filter). An M -state filter has transitions

$$i_{n+1} = U_n(i_n, g_{n+1}, x_{n+1}), \quad i_n \in \{1, \dots, M\}.$$

It reads only its previous index, the current command and current report. Past commands, past reports and retained command-generation or update seeds are unavailable outside the index. Fresh private coins may be used and then discarded. At a checkpoint the decoder reads only i_n and a newly revealed independent query. The known clock permits stage-dependent transition and output tables. Real command inputs and read-only real calibration constants are allowed; arithmetic workspace, execution time and finite-precision implementation of these fixed functions are not the memory resource of this definition. In particular, this is a finite-state transducer, not an algorithm that repeatedly rereads an exact prefix.

For each $1 \leq n < N$ fix a finite spanning probe list $H_{n,1}, \dots, H_{n,s_n^q}$ as in Proposition 2.3, using $m = N - n$ trials. To avoid confusion, s_n^q is the number of queries, not the exact state

s_n . After the checkpoint state is formed, choose an independent uniform query J . Announce $a \in [0, 1]$, execute that probe using its m fresh trials, and let Y be its all-failure indicator. All N trials are counted. The conditional squared-loss excess over the full-history predictor is

$$(7) \quad \mathcal{R}(a, p) = \frac{1}{s_n^q} \sum_j (a_j - p_j)^2, \quad p_j = \mu(P_h H_{n,j}).$$

This follows by expanding $\mathbb{E}[(a - Y)^2 \mid h, J]$; decoder randomization is averaged on the right. Only one selected checkpoint and one probe are executed in a run. Simultaneous bounds over checkpoints do not count the unexecuted probes as physical trials.

4. ATTAINABLE TRANSVERSALITY AND THE EXACT MEMORY PROFILE

We first prove the positivity statement that prevents a prior-dependent loss of rank. No density assumption on the prior is needed.

Lemma 4.1 (Strict mixed-moment positivity). *Let $\alpha_1 < \dots < \alpha_s$ and $\beta_1 < \dots < \beta_s$ be nonnegative real numbers. For every full-support probability on I ,*

$$(8) \quad \det[\mu(t^{\alpha_i + \beta_j})]_{i,j=1}^s > 0.$$

Consequently a rectangular moment matrix indexed by two increasing finite sets of distinct nonnegative exponents has rank the smaller cardinality.

Proof. For increasing positive points $t_1 < \dots < t_s$, the generalized Vandermonde determinant $\det[t_j^{\alpha_i}]$ is positive. Here is a self-contained sign argument. Put $t_j = e^{x_j}$. A nonzero combination of s exponentials with distinct ordered exponents has at most $s - 1$ distinct zeros: factor out the first exponential, differentiate, and use Rolle's theorem inductively. Thus the determinant cannot vanish on the connected region $x_1 < \dots < x_s$. When the x_j coalesce to x_0 from ordered nearby points, its first nonzero Taylor term has the sign of the Wronskian

$$e^{x_0 \sum_i \alpha_i} \prod_{i < j} (\alpha_j - \alpha_i)$$

times the ordinary positive Vandermonde in the x_j , divided by positive factorials. Its sign is therefore positive everywhere in that region. The same holds for the β determinant.

Expanding determinants and applying Fubini gives the finite determinant integration identity

$$\det[\mu(t^{\alpha_i + \beta_j})] = \frac{1}{s!} \int_{I^s} \det[t_j^{\alpha_i}] \det[t_j^{\beta_i}] \prod_{j=1}^s \mu(dt_j).$$

The integrand is nonnegative, by permutation symmetry and continuity at zero or repeated points. Choose s disjoint ordered subintervals in $(l, u) \cap (0, \infty)$. Each has positive prior mass by full support; on their product the integrand is strictly positive. The integral is strictly positive. Applying this to any square minor proves the last assertion. This is the classical total-positivity mechanism [7], with its needed form proved explicitly here. $\square$

Lemma 4.2 (A monomial tangent inside the attainable cube). *For every $n \geq 1$, choose distinct, sufficiently small positive numbers $c_1, \dots, c_n$. The failure factors $F_i = (1 + c_i t^D)/2$ are simultaneously attainable at interior commands. At this tuple the differential image of the unnormalized product map $W_A^n \rightarrow \text{span}\{t^s : s \in nA\}$ is*

$$(9) \quad T_{A,n} = \text{span}\{t^b : b \in B_{A,n}\},$$

where

$$(10) \quad B_{A,n} = \{jD : 0 \leq j \leq n\} \cup \bigcup_{a \in A \setminus \{0, D\}} \{a + jD : 0 \leq j < n\}.$$

In particular $\dim T_{A,n} = n(r-1) + 1$, and the product itself belongs to this tangent space.

Proof. Attainability follows by applying the fixed right inverse in Lemma 2.1 to $c_i t^D/2$ and taking all c_i small enough. The factors and their implementing commands lie in an open subset of the allowed cube. Constants 2^{-n} do not change a differential image, so omit them in the remaining calculation. Put $z = t^D$, $P(z) = \prod_i (1 + c_i z)$ and $P_i(z) = P(z)/(1 + c_i z)$. These n polynomials form a basis of the polynomials in z of degree at most $n-1$. Indeed evaluation at $z = -1/c_j$ makes all P_i except P_j zero, and $P_j(-1/c_j) \neq 0$.

The product differential is $\sum_i \dot{F}_i \prod_{j \neq i} F_j$. Variations in the constant direction give all polynomials in z through degree $n-1$; variations in t^D give their multiples by z . Together they give all z^j , $0 \leq j \leq n$. Variations in an interior exponent a give precisely $t^a z^j$, $0 \leq j < n$. The listed exponent sets are disjoint: two numbers $a + jD$ and $a' + j'D$ with $0 < a, a' < D$ agree only if $a = a'$ and $j = j'$, and none is a multiple of D . Their total cardinality is $(n+1) + n(r-2) = n(r-1) + 1$. Every summand of the differential has this form, proving equality rather than only inclusion in (9). The product is obtained by varying one factor along itself, so is in the tangent. $\square$

Theorem 4.3 (Generic normalized pairing rank). *For fixed $n, m \geq 1$, the map from n all-failure commands to the moment vector $(\mu(t^s P_h))_{s \in mA \setminus \{0\}}$ has generic rank $d_A(n, m)$ in (2). This holds for every full-support prior, including priors without a density. At an interior attainable tuple it has a smooth $d_A(n, m)$ -dimensional local image and a continuous local section to histories. A projection onto $d_A(n, m)$ of the moment or spanning-probe coordinates maps such a section onto an open Euclidean ball.*

Proof. Let $P = \prod_i F_i$, $Z = \mu P > 0$, and include the constant test in the linear map

$$L : T_{A,n} \longrightarrow \mathbb{R}^{h_A(m)}, \quad LQ = (\mu(t^s Q))_{s \in mA}.$$

At the tuple from Lemma 4.2, Lemma 4.1 gives

$$\text{rank } L = \min\{n(r-1) + 1, h_A(m)\}.$$

Let e_0 extract the coordinate of exponent zero, and put $p = L(P)/Z$. The derivative of the normalized prediction map on this tangent is

$$(11) \quad Q \longmapsto Z^{-1}(LQ - p e_0(LQ)).$$

Now $e_0 p = 1$ and $p \in L(T_{A,n})$. The linear map $v \mapsto v - p e_0 v$, restricted to $L(T_{A,n})$, has kernel exactly $\text{span}\{p\}$ and image in the zero-constant-coordinate hyperplane. It loses *exactly* one rank. Deleting that identically zero coordinate therefore gives rank $\min\{n(r-1), h_A(m) - 1\}$.

At every tuple the normalized product map is invariant under independent rescaling of its n nonzero factors, so its rank is at most $n(r-1)$. The number of nonconstant tests gives the other upper bound. Thus the attained rank is globally maximal. The command-to-factor linear map is surjective at every interior command; the same maximal rank is attained in command coordinates. Its Jacobian entries are rational functions of the commands with strictly positive polynomial denominator Z^2 . A maximal minor nonzero at the displayed tuple has a nonzero polynomial numerator. Its zero set is a proper real algebraic set with empty interior and zero Lebesgue measure. This proves genericity throughout the command cube, not only along the binomial family.

Choose $d = d_A(n, m)$ independent output coordinates and d command coordinates at that tuple, and freeze the complementary command coordinates. The inverse function theorem gives a local inverse for the selected output projection; shrink to a ball with compact closure. The full image of this section is a smooth local graph. A spanning list of probes is an injective linear image of the moment coordinates, so the same conclusion holds after choosing independent probe coordinates. All section histories are feasible all-failure histories. Their probability conditional on each parameter is at least η^n ; no failure evidence has been dropped. $\square$

Notice that when $d < h_A(m) - 1$ the attainable image can be curved. The proof does not replace it by its affine hull. The lower-bound ball is in a selected output projection, and its lift is a local graph, not necessarily a flat ball in the ambient prediction space.

Theorem 4.4 (Minimal continuous encoding and causal attainment). *The minimal dimension stated in Theorem 1.3 is $d_A(n, m)$. For a fixed N , there are compact exact state sets $\mathcal{S}_n \subset \mathbb{R}^{d_A(n, N-n)}$ and continuous-in-command causal updates attaining this dimension at every n . Their state represents the posterior-dependent future law, not an uncharged past-dependent task.*

Proof. Compose a sufficient continuous encoding with the section in Theorem 4.3. Different points in the projected ball have different predictions, so this composition is injective. A continuous injection of an open d -ball into $\mathbb{R}^k$, $k < d$, is impossible: embedding $\mathbb{R}^k$ as a coordinate plane of $\mathbb{R}^d$ would contradict invariance of domain. The decoder need not be continuous for this argument.

For the global upper bound, normalize each observed factor by its own prior integral, $\tilde{f}_i = f_i / \mu f_i$. Each normalized factor is in the affine hyperplane $\mu f = 1$ of W_A , with $r - 1$ coordinates. The chronologically ordered n factors determine the posterior product and use $n(r - 1)$ real coordinates. This representation may distinguish permutations; sufficiency does not forbid that. Alternatively, retain the $h_A(m) - 1$ nonconstant posterior moments in mA . Proposition 2.3 proves this alternative sufficient on every history. Use whichever representation is smaller. Both are continuous on each fixed-report stratum. Finite unions of compact continuous images of the command cube give compact state sets. The cases $n = 0$ or $m = 0$ have a single class and zero coordinates.

For causal implementation retain normalized factors while $n(r - 1) \leq h_A(N - n) - 1$. The left side increases and the right side nonincreases, so there is at most one changeover. When the inequality first fails, multiply the retained factors and compute the required moments. Afterward, if $m = N - n$ and the new report factor is $f(t) = \sum_{a \in A} f_a t^a$, update by

$$(12) \quad M'_s = \frac{\sum_{a \in A} f_a M_{s+a}}{\sum_{a \in A} f_a M_a}, \quad s \in (m - 1)A \setminus \{0\}, \quad M_0 = 1.$$

All indices $s + a$ belong to mA . The denominator is the report probability and is at least κ . At the changeover the integrands are products of n factor monomials and a remaining-test monomial, so their exponents are in $nA + (N - n)A = NA$. Normalizing each factor requires only moments in $A \subset NA$. Thus a finite read-only table $(\mu(t^s))_{s \in NA}$ suffices for all updates. Formula (12) then gives the exact shrinking state to time N . $\square$

For arbitrary nonmonomial spaces W , the starting identity remains (11), with the true product tangent and W^m . What is special here is the explicit attainable tangent (9) and the strictly positive mixed pairing. No dimension formula depending only on $\dim W$ is inferred outside this class.

5. COMPLETE HERMITE FLAGS AND FINITE SPECTRAL SCALES

We separate two classical coordinate facts from the attainable posterior argument. Throughout the confluent pairing, the scalar parameter is in $[0, 1]$ and μ is a fixed full-support probability measure. No density of this measure is required.

Lemma 5.1 (Arbitrary-prefix Newton–Hermite principle). *Let $y_1, \dots, y_p$ be any ordered multiset of nodes, with repetitions not necessarily adjacent. The functionals $\lambda_j f = [y_1, \dots, y_j]f$, $1 \leq j \leq p$, form a basis of the Hermite data at the full multiset: at each distinct node y , all derivatives of orders $0, \dots, m_y - 1$ are included, where m_y is its multiplicity. Consequently, for $H > 0$, the functions $[y_1, \dots, y_j](z \mapsto t^{Hz})$ span exactly*

$$\text{span}\{t^{Hy}(\log t)^k : y \text{ distinct, } 0 \leq k < m_y\}.$$

Proof. Every prefix divided difference is determined by the complete Hermite data at that prefix, so it belongs to the full Hermite data space. The latter has dimension p : a polynomial of degree less than p with all these data zero is divisible by $\prod_y (z - y)^{m_y}$ and hence is zero. On the polynomials $1, z, \dots, z^{p-1}$, λ_j annihilates degrees below $j - 1$ and takes value one on z^{j-1} . Its evaluation matrix is therefore upper triangular with diagonal one, also at repeated nodes by the Hermite extension. The p functionals are independent and span the data space. Substitution of t^{Hz} turns the k th derivative at y into $H^k t^{Hy}(\log t)^k$, with a nonzero constant factor. This proves the asserted function span. $\square$

This is the prefix interpolation principle of [12, Proposition 7, p. 48]. In particular, an arbitrary initial multiset gives complete multiplicity blocks, not isolated high logarithmic jets. The next mixed pairing is what connects those blocks to the product tangent of the experiment.

Lemma 5.2 (Strict confluent mixed pairing). *Let $\alpha_1 < \dots < \alpha_s$ be nonnegative, and let λ run through distinct nonnegative numbers with multiplicities q_λ summing to s . Include the columns $t^\lambda(\log t)^k/k!$, $0 \leq k < q_\lambda$, in this order within each increasing λ cluster. Assume $q_0 = 1$ if zero occurs. The matrix of their integrals against $t^{\alpha_i}\mu(dt)$ has strictly positive determinant. Consequently the corresponding rectangular pairing has full column rank whenever there are at least as many distinct row exponents as columns.*

Proof. Put $t = e^x$. The functions become $e^{\lambda x}x^k/k!$. A nonzero linear combination with total multiplicity s has at most $s - 1$ distinct zeros. Indeed divide by the first exponential and differentiate q_{λ_1} times. Rolle’s theorem reduces a hypothetical s zeros to at least $s - q_{\lambda_1}$ zeros. The remaining terms are exponential polynomials with distinct nonzero exponents and the same remaining polynomial degrees: the operator $(D + \lambda - \lambda_1)^{q_{\lambda_1}}$ is invertible on polynomials of each fixed degree bound. Induction gives a contradiction. If no remaining term occurs, the original function is an exponential times a polynomial of degree less than s , and the assertion is immediate.

Thus the evaluation determinant at distinct ordered points never vanishes. Its sign is positive. One can determine the sign by letting ordered simple exponents approach the prescribed repeated exponents and dividing by their positive within-cluster Vandermonde factors. Equivalently, the Wronskian of the displayed exponential-polynomial basis has sign

$$\prod_{\lambda < \lambda'} (\lambda' - \lambda)^{q_\lambda q_{\lambda'}} > 0;$$

the exponential factor and the factorial normalizations are positive. Nonvanishing fixes this sign throughout the connected ordered-point domain.

Use the determinant integration identity from Lemma 4.1, with one determinant replaced by this confluent evaluation determinant. Both determinants have the same positive sign on

ordered distinct positive points. The integrand is nonnegative elsewhere by symmetry and continuity. Separated ordered subintervals have positive prior mass, so the integral is strictly positive. The logarithmic functions are bounded at zero under the stated hypotheses, which also justifies Fubini. Select any s rows for the rectangular assertion. $\square$

Proposition 5.3 (Ordinary Vandermonde exterior scales). *Fix $q \geq 1$ and a multiset $x = (x_1, \dots, x_q) \in [0, 1]^q$. Let $W(x) = (x_i^{j-1})_{i,j=1}^q$, with singular values $\sigma_1 \geq \dots \geq \sigma_q \geq 0$, and let*

$$V_\ell(x) = \max_{|J|=\ell} \prod_{\{i,j\} \subset J} |x_i - x_j|, \quad V_1 = 1.$$

For $1 \leq \ell \leq q$, uniformly including repeated nodes,

$$(13) \quad V_\ell(x) \asymp_q \prod_{j=1}^{\ell} \sigma_j(W(x)) = \left\| \bigwedge_{j=1}^{\ell} W(x) \right\|.$$

Both sides vanish above the number of distinct sites. This is a comparison up to dimension-dependent constants, not an equality between a maximal minor and an exterior norm.

Proof. Permute rows into finite Leja order: choose a first site, then repeatedly maximize the product of distances to the previously chosen sites. Put $P_{j-1}(z) = \prod_{i < j} (z - x_i)$ and $d_j = |P_{j-1}(x_j)|$, with $d_1 = 1$. Every distance is at most one, so the greedy maximization and removal of candidates give $d_{j+1} \leq d_j$. If there are s distinct sites, the first s pivots are positive and all later pivots are zero.

Let T have as its j th column the coefficients of P_{j-1} in the power basis. It is upper triangular with diagonal one. Its entries are bounded elementary symmetric polynomials in numbers from $[0, 1]$. Thus $\|T\|$ is bounded in terms of q ; writing $T = I + U$ gives $T^{-1} = \sum_{k=0}^{q-1} (-U)^k$ and the same type of inverse bound. Evaluation gives $WT = N$, $N_{ij} = P_{j-1}(x_i)$. For $j \leq s$ put $L_{ij} = P_{j-1}(x_i)/d_j$. Greedy maximality bounds its entries in absolute value by one, and its leading s by s block is lower triangular with diagonal entries ± 1 . For $j > s$ set the j th column of $\tilde{L}$ to the coordinate vector e_j , retaining the first s columns of L . The resulting block lower triangular matrix and its inverse have norms bounded solely by q . The corresponding columns of N vanish, since their polynomials already vanish at every distinct site. Therefore

$$W = \tilde{L} \operatorname{diag}(d_1, \dots, d_q) T^{-1}.$$

Singular-value inequalities for left and right multiplication by an invertible matrix give $c_q d_j \leq \sigma_j(W) \leq C_q d_j$ for every j , including the zero tail.

Evaluation of the first ℓ monic Newton polynomials on any ℓ sites has determinant equal to their Vandermonde determinant. It is the selected-row submatrix of $L_{:,1:\ell}$ times $\operatorname{diag}(d_1, \dots, d_\ell)$ when $\ell \leq s$. The Leibniz formula bounds its first determinant by $\ell!$, while the first ℓ selected sites give absolute determinant one. Hence $\prod_{j \leq \ell} d_j \leq V_\ell \leq \ell! \prod_{j \leq \ell} d_j$. For $\ell > s$ both quantities vanish. Combining the comparisons proves (13); the equality with the exterior norm follows from the singular-value decomposition. $\square$

The preceding proof is the finite comparison in [23]. It uses no reciprocal gap and no hypothesis from a clustered Fourier matrix theorem. Applied to the formal future nodes it identifies the coordinate content of the scalar determinant profile. The attained subprobability cube, the global dimension-truncated cover, and the causal raw update are additional statements proved below; none follows merely from (13).

6. A GLOBAL DIMENSION-TRUNCATED COVERING ESTIMATE

We first record the real-geometric input with its anisotropic consequence. A bounded semialgebraic format means a bound on the number of variables, polynomial relations and their degrees in a defining formula; arbitrary real coefficients are allowed. Quantified formulas of fixed format can also be used, since real quantifier elimination gives a bound depending only on this format. Neither a coefficient norm nor a lower bound on a nonzero coefficient is part of the definition.

Lemma 6.1 (Bounded-format sets in a thin rectangle). *Let S be a nonempty compact semialgebraic subset of $\prod_{i=1}^q [-a_i, a_i]$, where $a_1 \geq \dots \geq a_q \geq 0$. Suppose its format is bounded and its dimension is at most p , with $1 \leq p \leq q$. There is a constant C depending only on the format and q such that, for every $\varepsilon > 0$,*

$$(14) \quad \mathcal{N}(S, \varepsilon) \leq C \sum_{j=0}^p \varepsilon^{-j} \prod_{i=1}^j a_i.$$

The empty product is one. Consequently S has a covering by at most M representatives belonging to S , of radius at most

$$(15) \quad C' \max_{1 \leq j \leq p} \left(M^{-1} \prod_{i=1}^j a_i \right)^{1/j}.$$

If all a_i vanish, one representative suffices with radius zero.

Proof. We use two classical real-geometric facts. First, intersections of a bounded-format semialgebraic set with arbitrary affine subspaces have a uniform bound on their number of connected components. The bound is independent of the coefficients; one formulation is the regularity statement in [18, Lemma 2.18], together with real quantifier elimination. Second, the real Vitushkin–Ivanov entropy inequality is

$$(16) \quad \mathcal{N}(S, \varepsilon) \leq C_q \sum_{j=0}^q \varepsilon^{-j} V_j(S), \quad V_j(S) = c_{q,j} \int b_0(S \cap L) \, dL.$$

For $j > 0$ the integral is over affine subspaces of codimension j , with normalized rotational measure and Lebesgue translation measure; $V_0(S) = b_0(S)$. This is the classical real inequality in [16, Theorem 3.5], also recalled as equations (4)–(5) in [17]. No nonarchimedean result is used here.

For $j > p$, almost every such affine section is empty: its translation parameter lies outside a projection of dimension at most $p < j$. Thus $V_j(S) = 0$. For $1 \leq j \leq p$, write an affine section as $L = u + E^\perp$, $u \in E$, for a j -plane E . If it meets S , then u belongs to the projection onto E of the containing rectangle. The component bound, denoted B , therefore gives

$$V_j(S) \leq c_{q,j} B \int_{G(q,j)} \text{vol}_j \left(\text{proj}_E \prod_i [-a_i, a_i] \right) dE.$$

This projected rectangle is a zonotope. Its volume is the sum, over j -element subsets of the coordinate directions, of their parallelepiped volumes. Each corresponding orthogonal-projection minor has absolute value at most one. Hence its volume is at most $2^j \binom{q}{j} a_1 \cdots a_j$. Also $V_0(S) \leq B$. Substitution into (16) proves (14), including zero-width axes. This argument explains why the products stop at the dimension of S , not at the number of sides of its ambient rectangle.

To respect the integer budget, let e be the maximum in (15) without its constant. If $a_1 > 0$, then $e > 0$ and $a_1 \cdots a_j \leq M e^j$ for every $j \leq p$. Choose $L \geq 2$ with $C \sum_{j=1}^p L^{-j} \leq 1/2$. For

$M \geq 2C$, (14) at radius Le is at most $C + M/2 \leq M$. Discard empty balls and replace each center by a point of S in its ball; the radius at most doubles. For $M < 2C$, a single point of S covers it at radius $2\sqrt{q}a_1$, while $e \geq a_1/M \geq a_1/(2C)$. Increasing the constant proves the assertion for these budgets as well. A nearest-representative map, with a fixed tie convention, is Borel. If $a_1 = 0$, the set is a singleton. $\square$

The bounded-format hypothesis is essential to a uniform statement of this kind. An unrestricted smooth image could develop arbitrarily many sheets. In the present experiment this hypothesis is a consequence of the finite product likelihood, not an extra regularity assumption on the prior or on an unknown attainable variety.

7. INTRINSIC RESOLUTION AT ARBITRARY EXPONENT COLLISIONS

We now prove the scalar resolution theorem. Maximal Vandermonde products on the future exponent set determine both attained checkpoint resolution and causal memory. They are defined at an exact collision without choosing a path into it. The finite spectral interpretation is Proposition 5.3; the argument here establishes attainability, the global image bound, and compatibility with causal updates.

7.1. Calibration chamber and determinant volumes. Fix $r \geq 2$ and a compact subset K of the open chamber

$$\{(a_1, \dots, a_{r-1}) : 0 < a_1 < \dots < a_{r-1}\}, \quad a_0 = 0.$$

The parameter of the experiment is $t \in [0, 1]$, with one fixed full-support prior μ . The known calibration is $a \in K$. One fixed matrix $C = (c_{ji})$ of column rank r defines the cells

$$k_j^a(t) = \sum_{i=0}^{r-1} c_{ji} t^{a_i}, \quad \sum_j k_j^a(t) = 1, \quad k_j^a(t) \geq k_* > 0.$$

The command cube is $[\eta, 1 - \eta]^J$, $0 < \eta < 1/2$, as before. These assumptions are imposed on all of K ; no regularity or semialgebraicity of K is required. Compactness inside the chamber separates the one-step exponents, but *does not* separate their additive sums. The finitely many report likelihoods have a common positive lower bound κ .

Fix $N \geq 2$, and choose once and for all $H \geq \max\{1, N \max_{a \in K} a_{r-1}\}$. For $m \geq 1$ write

$$\Lambda_m = \{\alpha \in \mathbb{N}_0^r : |\alpha| = m\}, \quad \Lambda_m^+ = \Lambda_m \setminus \{me_0\}, \quad q_m = |\Lambda_m^+|, \quad x_\alpha(a) = H^{-1} \sum_i \alpha_i a_i.$$

Here $\mathbb{N}_0$ includes zero. Formal labels are retained even when their values coincide. All x_α , $\alpha \in \Lambda_m^+$, lie in $[a_*/H, 1]$, where $a_* = \min_{a \in K} a_1 > 0$. For $1 \leq \ell \leq q_m$ define

$$(17) \quad \mathcal{V}_{m,\ell}(a) = \max_{\substack{J \subset \Lambda_m^+ \\ |J|=\ell}} \prod_{\substack{\alpha, \beta \in J \\ \alpha < \beta}} |x_\alpha(a) - x_\beta(a)|, \quad \mathcal{V}_{m,1} = 1.$$

The ordering inside the product is immaterial. The maximum is zero if fewer than ℓ distinct positive future exponents remain. Put

$$(18) \quad \begin{aligned} p_{n,m} &= \min\{n(r-1), q_m\}, \\ \Psi_{n,m}(M, a) &= \max_{1 \leq \ell \leq p_{n,m}} \left(\frac{\mathcal{V}_{m,\ell}(a)}{M} \right)^{2/\ell}, \\ \Xi_N(M, a) &= \max_{1 \leq n \leq N} \Psi_{n, N-n}(M, a). \end{aligned}$$

Set $\Psi_{0,m} = \Psi_{n,0} = 0$. Changing H to any other fixed admissible value changes only comparison constants. In particular the invariant does not depend on a convention for measuring exponents.

Use the failure-factor basis $1/2$ and $1/2 + \delta t^{a_i}$, $1 \leq i < r$, realized by the fixed right inverse of C for one sufficiently small $\delta > 0$. Its ordered m -fold products are the query menu, sampled uniformly after the memory index is formed. Expansion in the symbolic variables $X_0, \dots, X_{r-1}$ gives a fixed matrix G_m of column rank $|\Lambda_m|$: an invertible linear change of variables on homogeneous polynomials of degree m is invertible. Consequently the physical query norm and the norm of the formal raw moment vector are uniformly equivalent. Coincident exponent values impose relations on attainable vectors, not a loss of column rank of this *formal* coefficient matrix.

7.2. A uniformly conditioned flag. Order the positive nodes greedily. Choose the first label by a fixed rule; for example take its least node, breaking ties by a fixed label order. Having selected $x_1, \dots, x_{j-1}$, select a remaining label maximizing

$$\prod_{i < j} |x - x_i|, \quad d_1 = 1, \quad d_j = \prod_{i < j} |x_j - x_i|.$$

Break every tie by the same label rule. After all distinct values have been selected, continue through the remaining labels; their d_j are zero. This is finite Leja ordering. The ordering and its use in Newton interpolation are classical [20, 21]; the estimates needed here are proved below, including their zero-pivot form.

Lemma 7.1 (Newton scales and maximal determinant volumes). *For every calibration, $1 = d_1 \geq d_2 \geq \dots \geq d_{q_m} \geq 0$. There is a matrix L_a , whose entries have absolute value at most one, such that*

$$(19) \quad (t^{Hx_i})_{i=1}^{q_m} = L_a \operatorname{diag}(d_j) (g_j^a(t))_{j=1}^{q_m}, \quad g_j^a(t) = [x_1, \dots, x_j](z \mapsto t^{Hz}).$$

Divided differences at repeated nodes have their Hermite values. If s is the number of distinct positive nodes, the first s rows and columns of L_a form a lower triangular matrix with diagonal entries ± 1 ; its inverse has a bound depending only on q_m . Thus, for any vector v ,

$$(20) \quad c_m \|\operatorname{diag}(d_j)v\| \leq \|L_a \operatorname{diag}(d_j)v\| \leq C_m \|\operatorname{diag}(d_j)v\|.$$

Moreover, for every $1 \leq \ell \leq q_m$,

$$(21) \quad \prod_{j=1}^{\ell} d_j \leq \mathcal{V}_{m,\ell}(a) \leq \ell! \prod_{j=1}^{\ell} d_j.$$

All these assertions hold at exact coincidences.

Proof. Every distance is at most one. The maximization defining d_j and the removal of a candidate imply $d_{j+1} \leq d_j$. Put $P_{j-1}(z) = \prod_{i < j} (z - x_i)$. For $d_j > 0$ set $L_{ij} = P_{j-1}(x_i)/d_j$; for $d_j = 0$ set that column to zero. A previously selected node gives zero when $i < j$, and the greedy choice gives $|L_{ij}| \leq 1$ on all remaining nodes. The leading nonzero block is lower triangular with diagonal ± 1 . Back substitution bounds its inverse in a fixed dimension. If $d_j = 0$, all remaining node values already occur among the earlier ones, so P_{j-1} vanishes on every node. Newton's interpolation identity, evaluated at the nodes, is therefore exactly (19), also when there are repeats. Only the first s scaled coordinates can be nonzero, proving (20).

For $\ell \leq s$, evaluation of the monic polynomials $P_0, \dots, P_{\ell-1}$ on any ℓ chosen nodes has determinant equal to their Vandermonde determinant. Its matrix is the selected rows of $L_{:,1:\ell}$ times $\operatorname{diag}(d_1, \dots, d_\ell)$. The Leibniz formula bounds the first determinant by $\ell!$. For the first ℓ selected nodes its absolute value is one, giving both bounds in (21). For $\ell > s$, both sides vanish. No division by a zero pivot has been made. $\square$

The next assertion is the attainable part of the argument. Bounds for a Vandermonde matrix alone would not prove it.

Lemma 7.2 (Uniform attainable Newton flags). *For every $n, m \geq 1$ with $n + m \leq N$, let $z_a(h) = (\mathbb{E}[g_j^a(t) \mid h])_{j=1}^{q_m}$ and $p = p_{n,m}$. At one common interior all-failure command tuple, the first p coordinates of z_a have a surjective differential. Its least row singular value is bounded below uniformly over $a \in K$ and every ordering of the formal nodes, including limiting orderings at collisions. Under independent uniform acquisition commands, their all-failure subprobability law dominates β times uniform probability on a translated p -cube of side 2ρ , where $\beta, \rho > 0$ are uniform.*

Proof. We first justify continuity at all repeated-node configurations, without an assumption on how a collision is approached. The Hermite–Genocchi formula [12] expresses $g_j^a(t)$ as the integral of $(H \log t)^{j-1} t^{Hz}$ over the simplex of convex combinations of $x_1, \dots, x_j$; the simplex has volume $1/(j-1)!$. Since $Hx \geq a_* > 0$,

$$|g_j^a(t)| \leq \frac{H^{j-1}}{(j-1)!} t^{a_*} |\log t|^{j-1}, \quad 0 < t \leq 1.$$

The right side is bounded and tends to zero at $t = 0$. The same argument with higher logarithmic powers gives continuity, uniform in t , as the nodes vary in their compact range. Define $g_j^a(0) = 0$.

For any initial multiset of p nodes, the p divided-difference functionals are independent on polynomials of degree at most $p-1$: their values on $1, z, \dots, z^{p-1}$ form an upper triangular matrix with diagonal one. They span the Hermite evaluation functionals at the distinct nodes, with all derivative orders below the corresponding multiplicities. Applied to t^{Hz} , their span is therefore the complete exponential-polynomial space

$$\text{span}\{t^{H\lambda}(\log t)^k : \lambda \text{ in the multiset, } 0 \leq k < m_\lambda\}.$$

This argument permits repeated nodes at nonadjacent positions. Adjoin the constant function. The resulting $p+1$ tests form a complete confluent Chebyshev space, not a selection of isolated high derivatives.

Choose the common binomial factors $F_i = (1 + c_i t^{a_{r-1}})/2$, with distinct sufficiently small positive c_i . Their command coefficients are independent of a . By Lemma 4.2 their product tangent has $n(r-1) + 1$ distinct monomials. The mixed and confluent pairing of Lemmas 4.1 and 5.2 has rank $p+1$ against the selected tests, at every node configuration. In the normalized differential

$$Q \mapsto (\mu F)^{-1}(LQ - v e_0 LQ), \quad v = LF/\mu F,$$

the product belongs to the tangent and $e_0 v = 1$. Thus exactly the constant direction is lost and the differential onto the other p coordinates is surjective.

For each fixed permutation of formal labels this derivative is continuous on K , including coincidences, by the uniform bounds just proved and positivity of evidence. There are finitely many permutations. Compactness therefore supplies a common positive least row singular value. First and second derivatives in command coordinates are uniformly bounded because the numerator is a polynomial and the evidence is at least κ^n .

To record the probability step explicitly, complete the derivative's row space by orthonormal kernel coordinates. A ball of uniform radius in command space gives a square map whose derivative differs from its invertible central derivative by at most half its least singular value. The contraction proof of the inverse theorem gives a fixed-size product box in output and kernel coordinates, with its inverse inside the command cube. The inverse Jacobian determinant is bounded below by the reciprocal of a uniform power of the forward derivative bound. The command density is constant, and the joint density of those commands and the all-failure word

is at least that density times η^n . Integrating over the kernel box proves the stated subprobability minorization. No prior density is involved and no failure evidence is discarded. $\square$

7.3. The global attainable law.

Theorem 7.3 (Collision-uniform checkpoint classification). *Under the assumptions of this section there are $0 < c \leq C < \infty$ such that, for every $a \in K$ and integer $M \geq 1$,*

$$(22) \quad c\Psi_{n,m}(M, a) \leq \bar{R}_{M;n,m}^a \leq R_{M;n,m}^{a,\max} \leq C\Psi_{n,m}(M, a).$$

The unconditional lower law uses independent uniform acquisition commands. The upper code is deterministic, has at most M reachable representatives, and works on every admitted history. The lower law permits the independent coding randomization of Definition 3.1. Constants may depend on K, C, η, μ, n, m , but not on a or M .

Proof. Let $D_a = \text{diag}(d_j)$. For one report word each coordinate of $z_a(h)$ has the form $N_j(u)/Z(u)$, with numerator and evidence polynomials of degree at most n in the command variables and $Z(u) \geq \kappa^n$. Thus

$$S_a = \{D_a z_a(h) : |h| = n\}$$

is compact and has coefficient-independent bounded semialgebraic format: it is a finite union of projections of the formulas $Z(u)y_j = d_j N_j(u)$ on the command cube. Prior integrals and node values are real coefficients in these formulas. They need not depend semialgebraically on a .

Individually normalizing the n report factors puts them in affine hyperplanes of dimension $r - 1$. The moment map is rational in those coordinates. Hence $\dim S_a \leq p$. Lemma 7.2 also gives $|z_{a,j}| \leq B$, so S_a lies in the rectangle with half-widths $Bd_1 \geq \dots \geq Bd_{q_m}$. By Lemma 6.1, it has at most M reachable representatives with covering radius bounded by

$$C \max_{\ell \leq p} \left(M^{-1} \prod_{j \leq \ell} d_j \right)^{1/\ell}.$$

The physical metric equivalence follows from (20) and the full-column-rank product-probe matrix G_m . A fixed tie rule makes nearest-representative encoding Borel. Squaring and applying (21) gives the upper bound in (22), with the integer budget already enforced by Lemma 6.1.

For the lower bound, first average any randomized decoder conditional on its label; squared loss cannot increase. Orthogonally project its query centers onto the affine physical query span, apply a fixed left inverse of G_m , and then the inverse of the leading nonzero block of L_a . These maps have uniformly bounded norms. Projection onto any first $\ell \leq \min(p, s)$ coordinates reduces the squared-error problem to the marginal of the cube in Lemma 7.2, scaled by $d_1, \dots, d_\ell$. The probability that M radius- b balls cover this uniform rectangle is at most

$$C_\ell M b^\ell / (d_1 \cdots d_\ell).$$

Choose $b = c_\ell (d_1 \cdots d_\ell / M)^{1/\ell}$ with the constant small enough that at least half the minorized mass remains. Its mean squared error is at least a fixed multiple of b^2 . An encoder's randomized assignment cannot improve upon the distance to the nearest of its centers. Independent public coding randomness can be fixed and then averaged, since it does not generate the acquisition law. Maximize over ℓ . For $\ell > s$ the determinant volume is zero, so no additional term is needed. Equation (21) now gives the lower bound. The unconditional optimum is bounded above by the minimax optimum, completing the proof. $\square$

Theorem 7.4 (Causal classification across all collision strata). *For the same fixed experiment and horizon, there exist $c, C > 0$ such that for every known $a \in K$ and every integer $M \geq 1$ there exists a single, stage-compatible deterministic M -state transducer with maximum-history,*

maximum-checkpoint regret at most $C\Xi_N(M, a)$. Every M -state transducer has maximum unconditional checkpoint regret at least $c\Xi_N(M, a)$ under one common independent uniform exploration law. Equivalently,

$$(23) \quad c\Xi_N(M, a) \leq \mathfrak{R}_{M,N}^{a,\text{av}} \leq \mathfrak{R}_{M,N}^{a,\text{max}} \leq C\Xi_N(M, a).$$

The transducer may depend on a and M . Constants are uniform in both; codebooks are not asserted to be calibration-blind. It stores only its representative index. The clock, known calibration and fixed real-valued program are read-only, as in Definition 3.1.

Proof. At stage n let $m = N - n$ and store a representative in the reachable raw moment image indexed by Λ_m , including the constant coordinate $v_{me_0} = 1$. For a new factor $f(t) = \sum_i f_i(g, x)t^{a_i}$, the exact update is

$$(24) \quad v'_\alpha = \frac{\sum_i f_i(g, x)v_{\alpha+e_i}}{\sum_i f_i(g, x)v_{e_i+(m-1)e_0}}, \quad \alpha \in \Lambda_{m-1}.$$

Every index on the right belongs to Λ_m . Coincident labels have equal moment values and cause no ambiguity. On the segment joining any two reachable moment vectors, the denominator is a report probability under a posterior mixture and is at least κ . Coefficients are uniformly bounded and raw moments belong to $[0, 1]$. Differentiating the quotient gives a Lipschitz constant independent of all exponent gaps.

At every stage Theorem 7.3 provides a raw reachable codebook of cardinality at most M and covering radius at most $C_n\Psi_{n,N-n}(M, a)^{1/2}$. Update a representative by (24), then quantize in the next reachable codebook. Updating a reachable representative produces a reachable next state, so this quantizer is always evaluated on its domain. If e_n is the raw error, then

$$e_0 = 0, \quad e_{n+1} \leq L_n e_n + C_{n+1}\Psi_{n+1,N-n-1}(M, a)^{1/2}.$$

The finite recurrence bounds every e_n by $C_N\Xi_N(M, a)^{1/2}$. The query matrix converts this to the asserted squared-regret bound. The exact prefix appears only in this error analysis, not in the transducer. No divided-difference coordinate, reciprocal pivot or stored command tape is used in a transition.

Conversely, at each checkpoint the streaming index is an M -message encoder of the prefix. All command laws used in Theorem 7.3 are prefixes of one common exploration law. Each transducer therefore satisfies every checkpoint lower bound; taking their maximum and then the infimum proves the lower bound in (23). Finitely many checkpoint constants can be replaced by their positive minimum. One checkpoint and its remaining query trials are executed in a run, as before. $\square$

Corollary 7.5 (Intrinsic bit law). *Let $B_*(\varepsilon, a)$ be the least integer persistent bit budget for maximum-history streaming regret at most ε in this experiment. Uniformly for $0 < \varepsilon \leq 1$ and $a \in K$,*

$$(25) \quad B_*(\varepsilon, a) = \max_{\substack{1 \leq n < N \\ 1 \leq \ell \leq p_{n,N-n}}} \left[\frac{\ell}{2} \log_2 \frac{1}{\varepsilon} + \log_2 \mathcal{V}_{N-n,\ell}(a) \right]_+ + O(1).$$

A zero determinant volume contributes zero to this maximum, by interpreting its logarithm as $-\infty$ before taking the positive part. The same law holds for the unconditional maximum-checkpoint criterion. At each fixed calibration the small-error slope is $D_A(N)/2$.

Proof. With $M = 2^B$, solving each inequality $(\mathcal{V}_{m,\ell}/M)^{2/\ell} \leq \varepsilon$ gives the displayed lower bound on B . Constants in (23) change it by at most a bounded amount, since ℓ is bounded. Integer rounding costs at most one further bit. For fixed a , the largest nonzero index at checkpoint (n, m) is $\min\{n(r-1), |mA| - 1\} = d_A(n, m)$, proving the slope assertion. $\square$

7.4. Collision orders and a finite tree formula. The determinant form has a consequence not requiring interpolation coordinates: it classifies arbitrary orders of tangency to a collision. It also gives a finite optimization rule for all resolution phases.

Theorem 7.6 (Collision-tree determination of memory phases). *Let $a(\theta) \in K$, $0 \leq \theta \leq \theta_0$, be a known path. For a fixed m , identify formal sums that are identical along the path. Suppose that for every remaining pair of labels*

$$|x_\alpha(\theta) - x_\beta(\theta)| \asymp \theta^{w_{\alpha\beta}}, \quad w_{\alpha\beta} \in [0, \infty), \quad 0 < \theta \leq \theta_0.$$

After decreasing θ_0 this hypothesis holds for every real analytic path whose nonidentical sums are analytic at zero. Define

$$(26) \quad \beta_{m,\ell} = \min_{\substack{J \text{ a set of} \\ |J|=\ell}} \sum_{\{\alpha,\beta\} \subset J} w_{\alpha\beta}, \quad \beta_{m,1} = 0.$$

Then $\mathcal{V}_{m,\ell}(a(\theta)) \asymp \theta^{\beta_{m,\ell}}$. For unavailable cardinalities the volume is zero. Consequently the checkpoint law is

$$(27) \quad \overline{R}_{M;n,m}^{a(\theta)} \asymp \max_{\ell \leq p_{n,m}} \theta^{2\beta_{m,\ell}/\ell} M^{-2/\ell},$$

with zero-volume terms omitted; the streaming law is its maximum over checkpoints. The exponents $\beta_{m,\ell}$ depend only on the pairwise collision orders, and can be computed by a finite tree allocation rule.

Proof. Each fixed subset product has order equal to the sum of its pair orders. There are finitely many subsets, so their maximum has the smallest such order. This proves the determinant assertion and then (27) by Theorems 7.3 and 7.4. Constants may depend on the path's nonzero leading coefficients; the intrinsic determinant theorem, in contrast, is already uniform on K .

For completeness we give the allocation rule and its justification. The triangle inequality for node differences implies

$$w_{\alpha\gamma} \geq \min\{w_{\alpha\beta}, w_{\beta\gamma}\}.$$

Indeed a smaller order on the left would contradict the upper bound by the sum of the two differences on the right as $\theta \downarrow 0$. Thus thresholds on w define nested equivalence classes. Make a rooted tree from these classes, with root height zero; an internal cluster C has height $h_C = \min_{\alpha \neq \beta \in C} w_{\alpha\beta}$, and its children are the equivalence classes for orders strictly greater than h_C . Split the root at orders greater than zero. Singleton children are leaves. Redundant equal-height vertices can be omitted. For an internal nonroot vertex put $\Delta_C = h_C - h_{\text{parent}(C)}$. The sum of these increments on the common ancestor chain of two leaves is exactly their pair order. Hence, for a selected leaf set J ,

$$(28) \quad \sum_{\{\alpha,\beta\} \subset J} w_{\alpha\beta} = \sum_{\substack{C \text{ internal,} \\ C \neq \text{root}}} \Delta_C \binom{|J \cap C|}{2}.$$

Set $F_C(0) = F_C(1) = 0$ for a leaf and $F_C(k) = +\infty$ for $k > 1$. For an internal vertex, including the root with $\Delta_{\text{root}} = 0$, compute

$$(29) \quad F_C(k) = \Delta_C \binom{k}{2} + \min_{\sum_D \text{ child of } C} \sum_{k_D=k} F_D(k_D).$$

Equation (28) proves this recurrence by induction. It gives $F_{\text{root}}(\ell) = \beta_{m,\ell}$. Successive finite min-convolutions compute the table; no optimization over measures, posterior encoders or codebooks is involved. For analytic paths the pair orders are the finite orders of their first nonzero Taylor coefficients, after identical sums have been identified. This proves the assertion. $\square$

For the affine family of Section B, different limiting clusters have pair order zero, and distinct members of a cluster have pair order one. Thus a choice of k nodes from that cluster costs $\binom{k}{2}$; its successive costs are $0, 1, \dots, s-1$. Minimizing their sum picks the smallest costs from all clusters. Therefore $\beta_{m,\ell} = \nu_{m,1} + \dots + \nu_{m,\ell}$. This recovers Theorem D.1, while its separate proof remains available. Higher orders of contact between distinct branches require the pair orders above and are not encoded by the affine multiplicity list.

7.5. Intersecting collision loci and arbitrary tangency order. Here is a two-parameter consequence with different ranks on intersecting loci and arbitrarily deep resolution phases along tangent approaches. It illustrates the geometry of the determinant theorem rather than requiring a separate asymptotic analysis for each path.

Corollary 7.7 (A two-parameter collision arrangement). *Let $A_{u,v} = \{0, 1, 2+u, 3+v\}$, with $|u|, |v| \leq 1/16$, and take the four cells*

$$k_0(t) = \frac{1}{4} + \frac{1}{16}(t + t^{2+u} + t^{3+v}), \quad k_i(t) = \frac{1}{4} - \frac{1}{16}t^{a_i}, \quad 1 \leq i \leq 3.$$

Use any fixed admitted command cube and a fixed full-support prior. For total horizon five set $\rho = \max\{|u|, |v|\}$ and $\tau = \min\{|u|, |v-u|, |v-2u|\}$. Then, uniformly on the square,

$$(30) \quad \mathfrak{R}_{M,5}^{u,v} \asymp \max \left\{ M^{-1/3}, \rho^{1/2} M^{-1/4}, (\rho^2 \tau)^{2/9} M^{-2/9} \right\}$$

for either streaming criterion. The exact peak dimension is nine off the three collision lines, eight on any such line away from their intersection, and six at $(0,0)$.

Along $u = \theta$, $v = \theta + \theta^k$ with an integer $k \geq 2$, the three regimes have respective orders

$$M^{-1/3}, \quad \theta^{1/2} M^{-1/4}, \quad \theta^{(4+2k)/9} M^{-2/9}.$$

Their crossover budgets have orders θ^{-6} and $\theta^{-(8k-2)}$, with regrets of orders θ^2 and θ^{2k} , respectively. These are comparison orders, not exact integer thresholds.

Proof. The cells sum to one, lie between $3/16$ and $7/16$, and their coefficient matrix has rank four: its last three rows separate the three nonconstant coefficients and the sum of its rows gives the constant. They therefore satisfy the compact-chamber hypotheses.

At checkpoint $n = 3, m = 2$ the nine positive formal exponents are

$$1, \quad 2, \quad 2+u, \quad 3+u, \quad 3+v, \quad 4+2u, \quad 4+v, \quad 5+u+v, \quad 6+2v.$$

They form six uniformly separated groups, with three within-group gaps $|u|, |v-u|, |v-2u|$. Denote these gaps in decreasing order by $g_1 \geq g_2 \geq g_3$. For $\ell \leq 6$ the determinant volumes are comparable to one. For $\ell = 6+j$, $1 \leq j \leq 3$, they are comparable to $g_1 \cdots g_j$: choose one exponent from every group and a second from the j groups with largest gaps for the lower bound. Any selection of $6+j$ exponents must double at least j groups; all other factors are bounded, proving the upper bound. Fixed powers of H only change constants. Since $p_{3,2} = 9$, its profile is

$$\max \{ M^{-1/3}, g_1^{2/7} M^{-2/7}, (g_1 g_2)^{1/4} M^{-1/4}, (g_1 g_2 g_3)^{2/9} M^{-2/9} \}.$$

The signed differences satisfy $(v-u) - (v-2u) = u$, so the largest absolute gap is at most twice the second largest. Moreover $g_1 \asymp \rho$, hence $g_2 \asymp \rho$ and $g_3 = \tau$. The seven-dimensional term is the geometric interpolation, with weights $3/7$ and $4/7$, of the six- and eight-dimensional terms, up to uniform constants. This proves the expression in (30) for the checkpoint, also on the collision lines.

At the other checkpoints, the attainable dimensions are bounded by three, six and three, respectively. Since every normalized gap is at most one, their squared covering errors are at most $CM^{-1/3}$. The causal classification therefore gives the same entire-horizon profile and the

checkpoint-three lower bound supplies its converse. Counting distinct two-fold exponents gives the three exact peak dimensions asserted.

Finally, on the stated path $\rho \asymp \theta$ and $\tau \asymp \theta^k$. Substitution gives the three displayed terms. Equality in order of the first two gives $M = \theta^{-6}$, at regret θ^2 ; equality in order of the last two gives $M = \theta^{-(8k-2)}$, at regret θ^{2k} . As $k > 1$, these budgets are strictly separated as $\theta \downarrow 0$ and all three regimes occur. The exponent k records contact with a collision locus, not an additional assumed observable jet. $\square$

8. SHARED MEMORY FOR INDEPENDENT EXPERIMENTS

The scalar classification is stable under taking independent products, although the corresponding future query space is a tensor product. The persistent resource is one joint index. In particular, the converse below does not assume a separate encoder, a product codebook, or a separate memory budget for each experiment.

8.1. The product experiment. Fix $B \geq 1$. For each $b \in \{1, \dots, B\}$ take an experiment satisfying Section 7, with exponent set A_b , number of one-step exponents r_b , compact calibration set K_b , coefficient matrix C_b , command cube, full-support prior μ_b , and horizon $N_b \geq 2$. These data, including B , are fixed. The hidden parameter is $t = (t_1, \dots, t_B) \in [0, 1]^B$ with prior $\mu = \bigotimes_{b=1}^B \mu_b$. Conditional on t , all raw trials are independent. A trial assigned to block b uses only that block's detector and current command. The known calibration is $a = (a^{(1)}, \dots, a^{(B)}) \in \prod_b K_b$.

A deterministic schedule ω is a word of length $T = \sum_b N_b$ containing N_b copies of b . After s acquisition trials let $n_b(s)$ be the number assigned to b and put $m_b(s) = N_b - n_b(s)$. At that checkpoint sample independently and uniformly one member of each block's ordered m_b -fold product probe menu. A block with $m_b = 0$ contributes the constant factor one. The selected probe executes all $T - s$ remaining trials, in the residual schedule order. Its target Y is the indicator that every one of those reports is a failure. All trials are consumed, including after a nonfailure. Thus s acquisition trials and the selected query together always use exactly T trials. There is one selected checkpoint per run, not simultaneous execution of all the counterfactual queries.

We use the squared-loss excess relative to the full-history Bayes predictor. The decoder receives the query tuple only after the acquisition index has been formed. Its persistent memory has at most M values in total, with the resource convention of Definition 3.1. The schedule and clock are read-only; previous commands and reports are not. For the unconditional criterion all acquisition commands, in every block and at every time, are independent and uniform on their respective cubes, independently of the hidden parameter and coding randomization. This is a single acquisition law whose prefixes serve all checkpoints.

Write $\bar{R}_{M;s,\omega}^a$ and $R_{M;s,\omega}^{a,\max}$ for the optimal unconditional and maximum-history checkpoint regrets. Write $\mathfrak{R}_{M,\omega}^{a,\text{av}}$ and $\mathfrak{R}_{M,\omega}^{a,\max}$ for the infima over one stage-compatible filter of, respectively, its maximum unconditional checkpoint regret and its maximum regret over checkpoints and histories. A checkpoint encoder may inspect its full prefix once; a streaming transition may not.

For $n, m > 0$, let $q_{b,m}$ and $\mathcal{V}_{b,m,\ell}$ be the positive formal label count and determinant volumes of block b . Put

$$p_b(n, m) = \min\{n(r_b - 1), q_{b,m}\}.$$

Set $p_b(n, m) = 0$ when $n = 0$ or $m = 0$. At a fixed checkpoint abbreviate $p_b = p_b(n_b, m_b)$ and define

$$(31) \quad V_{b,0} = 1, \quad V_{b,\ell} = \mathcal{V}_{b,m_b,\ell}(a^{(b)}) \quad (1 \leq \ell \leq p_b), \quad P_s = \sum_b p_b.$$

A block not yet observed, as well as a completed block, has one reachable future state and contributes only $V_{b,0}$. Formal positive labels are not discarded at collisions; zero volumes are allowed. Define

$$(32) \quad \begin{aligned} A_{s,k}(a) &= \max_{\substack{0 \leq \ell_b \leq p_b \\ \sum_b \ell_b = k}} \prod_{b=1}^B V_{b,\ell_b}, & A_{s,0} &= 1, \\ \Theta_{s,\omega}(M, a) &= \max_{1 \leq k \leq P_s} (A_{s,k}(a)/M)^{2/k}, \\ \Theta_\omega(M, a) &= \max_{0 \leq s \leq T} \Theta_{s,\omega}(M, a). \end{aligned}$$

An empty maximum for the checkpoint profile is zero. The dependence of $A_{s,k}$ on the schedule is through the count vector $(n_b(s))$. The operation in the first line is associative: grouping several blocks only groups the same finite maximization over allocations. We shall need its full finite-error form, not just its largest nonzero index.

8.2. Tensor queries retain the component metric. For block b , let v_b be its formal raw moment vector indexed by Λ_{b,m_b} , including the constant entry $v_{b,m_b e_0} = 1$. For $m_b = 0$ this vector is (1). Let $\mathcal{S}_b$ be its complete reachable image at the checkpoint. If $n_b = 0$, it is the singleton given by the known prior. All its entries lie in $[0, 1]$.

Lemma 8.1 (Product attainment and the tensor metric). *The joint posterior after an admitted history is the product of its block posteriors, and the set of attainable raw tuples is exactly $\prod_b \mathcal{S}_b$. If $Q(v)$ is the vector of probabilities of the full tensor query menu, then uniformly in calibration,*

$$(33) \quad c \sum_b \|v_b - \tilde{v}_b\|^2 \leq \|Q(v) - Q(\tilde{v})\|^2 \leq C \sum_b \|v_b - \tilde{v}_b\|^2.$$

There is, moreover, a bounded linear map on the entire query-coordinate space that takes $Q(v)$ to the concatenated vector $(v_b)_b$. The bound for this map is independent of exponent gaps.

Proof. Group the likelihood factors of a realized history by block, writing F_b for their product. Its evidence is $\int \prod_b F_b(t_b) \otimes_b \mu_b(dt_b) = \prod_b \mu_b F_b$. Bayes normalization consequently gives the product posterior. Every tuple of admitted block histories can be interleaved in the prescribed schedule; every report word has positive evidence. This proves the statement about the whole attainable image, not only a local family. The argument also applies to commands chosen using past reports: for each fixed realized history the chosen likelihood factors are known. The lower law, however, uses the independent commands already specified, not an adaptive command law.

Let G_b be the fixed, full-column-rank formal coefficient matrix for the block's product probes, and take $G_b = (1)$ for $m_b = 0$. Independence gives

$$Q(v) = G \bigotimes_b v_b, \quad G = \bigotimes_b G_b.$$

Each G_b is independent of calibration, hence G has a bounded fixed left inverse $G^\dagger$. In the tensor $\bigotimes_b v_b$, selecting entries in which only the b th index varies and every other index is the constant label gives precisely v_b . Let E be the concatenation of these coordinate-selection maps. Then $EG^\dagger Q(v) = (v_b)_b$, proving the last assertion and the lower inequality. The repeated constant entries cause no difficulty. For the upper inequality expand the difference of the two tensor products as a telescoping sum, changing one factor at a time. Every other factor has norm bounded by the square root of its fixed coordinate count. Cauchy–Schwarz and the fixed bound on G give the upper inequality. All constants are finite-dimensional; none contains a reciprocal exponent separation. $\square$

Although $Q(\prod_b \mathcal{S}_b)$ lies in a nonlinear tensor-product image, its metric is thus the component product metric. In particular, its tensor ambient dimension is not the attainable information dimension. The existence of the linear map on arbitrary query centers will also be needed for a converse against unrestricted joint encoders.

Theorem 8.2 (Exact information dimension under composition). *At checkpoint s , the smallest Euclidean dimension of a sufficient history encoding continuous on each report stratum is*

$$(34) \quad d_\omega(s, a) = \sum_{b=1}^B \min\{n_b(s)(r_b - 1), |m_b(s)A_b| - 1\}.$$

Sufficiency is for every common future-only continuation in the prescribed remaining block schedule. These dimensions are attained simultaneously by a causal exact-real state. Its least possible peak size is $D_\omega(a) = \max_s d_\omega(s, a)$.

Proof. Each realized future word has a likelihood in the tensor product of the block future observation spaces. Equality of the block future moments therefore implies equality of every such word probability under a common feedback continuation; integrating independent feedback randomization preserves that assertion. Conversely equality for the tensor probe menu recovers each block future vector by Lemma 8.1. Thus the quotient on attainable histories is the product of the scalar operational quotients.

For each active block, Theorem 4.3 supplies a smooth attainable patch of dimension $d_b = \min\{n_b(r_b - 1), |m_b A_b| - 1\}$ and a continuous section to all-failure histories. The product of these sections, interleaved by ω , is a continuous section of a $\sum_b d_b$ -dimensional patch. A sufficient continuous encoding is injective on this patch. Invariance of domain, applied after embedding a smaller target Euclidean space in $\mathbb{R}^{\sum_b d_b}$, excludes a smaller dimension. Singleton factors contribute zero.

For the upper bound run the exact causal construction of Theorem 4.4 separately in each block at its own local clock, concatenating its coordinates in the single state. On a trial of block b , update only that component. Before its first observation its state is known, and after its last trial it has no remaining future task. Each component changes from normalized factors to future moments at most once. Its coordinate count at every checkpoint is exactly d_b , so the concatenated count is (34). All needed block prior moments have exponents in $N_b A_b$. The lower bound holds at every checkpoint of any exact causal encoder, and hence also at the checkpoint maximizing the sum. This proves the peak assertion. As in the scalar theorem, an encoding of histories is not asserted to be a global coordinate chart of a singular quotient. $\square$

8.3. A joint converse and a globally attainable cover.

Theorem 8.3 (Shared-state composition law). *For the fixed block experiments there exist $0 < c \leq C < \infty$ such that, for every calibration $a \in \prod_b K_b$, every deterministic schedule ω , every checkpoint s and integer $M \geq 1$,*

$$(35) \quad c\Theta_{s,\omega}(M, a) \leq \bar{R}_{M;s,\omega}^a \leq R_{M;s,\omega}^{a,\max} \leq C\Theta_{s,\omega}(M, a).$$

Furthermore,

$$(36) \quad c\Theta_\omega(M, a) \leq \mathfrak{R}_{M,\omega}^{a,\text{av}} \leq \mathfrak{R}_{M,\omega}^{a,\max} \leq C\Theta_\omega(M, a).$$

The upper bound is attained by one deterministic transducer with at most M joint labels after every report. The lower bound permits arbitrary joint encoding and decoding, including the independent coding randomness of Definition 3.1. Constants may depend on the fixed block data and horizons, but not on a , M or ω . The filter may depend on all three. No product-code restriction is imposed on the optimum, and no extra persistent block indices are supplied.

Proof. Fix a checkpoint with at least one active block. For each active b , write $d_{b,1} \geq \dots \geq d_{b,q_b,m_b} \geq 0$ for its Leja scales, and put

$$D_{b,0} = 1, \quad D_{b,\ell} = \prod_{j=1}^{\ell} d_{b,j} \quad (1 \leq \ell \leq p_b).$$

Lemma 7.1 gives $D_{b,\ell} \leq V_{b,\ell} \leq \ell! D_{b,\ell}$, including zero values. Every active block has $D_{b,1} = 1$.

Upper bound on the entire product image. The bounded-format image and dimension argument in the proof of Theorem 7.3, followed by Lemma 6.1, gives at every radius $e > 0$ a cover of the block raw image with at most

$$C_b \sum_{\ell=0}^{p_b} D_{b,\ell} e^{-\ell}$$

balls of radius e , after changing a uniform constant for the raw metric. A singleton block needs only one ball. Taking products of these covers at radius $e/\sqrt{B}$ gives, in the concatenated metric, a cover bounded by

$$(37) \quad \mathcal{N} \left(\prod_b \mathcal{S}_b, e \right) \leq C_0 \sum_{k=0}^{P_s} A_{s,k}(a) e^{-k}.$$

Indeed expansion of the product sums over all bounded allocations (ℓ_b) ; there are only finitely many of them, and each product is bounded by its maximizing allocation in (32). The factors $B^{k/2}$ and the factorial comparisons are absorbed into C_0 . This argument keeps *each* past cap p_b . Applying a single total-dimensional rectangle estimate to the concatenated ambient box instead could wrongly allocate more than p_b axes to a block.

Set $e_* = \max_{1 \leq k \leq P_s} (A_{s,k}/M)^{1/k}$. For $M \geq 2C_0$, choose a constant L so large that $C_0 \sum_{k=1}^{P_s} L^{-k} \leq 1/2$. Substituting Le_* in (37) yields at most M balls. Discard empty balls and replace each remaining center by a reachable tuple in that ball, increasing radii by at most two. For $M < 2C_0$, a single reachable tuple covers the bounded product image at a fixed radius; since $A_{s,1} = 1$, $e_* \geq 1/M \geq 1/(2C_0)$, so a larger uniform constant suffices. The finite code thus respects the integer budget for every M . Lemma 8.1 transfers the cover to all tensor query predictions. Squaring its radius and dividing by the fixed menu size gives the checkpoint upper bound. A fixed nearest-center tie rule makes the encoder Borel.

Lower bound against joint centers. Take any allocation (ℓ_b) with $k = \sum_b \ell_b \geq 1$ and $\prod_b V_{b,\ell_b} > 0$. Under the common exploration law the block histories are independent. For each block with $\ell_b > 0$, project the minorized p_b -cube of Lemma 7.2 onto its first ℓ_b coordinates and scale them by $d_{b,1}, \dots, d_{b,\ell_b}$. The product of the corresponding all-failure subprobability laws dominates a fixed positive multiple of uniform probability on a translated k -rectangle of volume

$$c_1 \prod_b D_{b,\ell_b}.$$

Blocks with $\ell_b = 0$ are integrated out; no density assumption on any μ_b is used. The positive multiplier includes the actual acquisition evidence in all selected blocks. Constants can be chosen uniformly over all allocations and count vectors because these are finite in number.

For an arbitrary decoder, average its private randomness conditional on the index and query. This leaves at most M centers in the entire query vector space and cannot increase squared loss. Apply to those centers the linear map $EG^\dagger$ of Lemma 8.1, then the inverse of each active Leja block and the selected-coordinate projections. These maps recover exactly the scaled Newton coordinates of a true query vector, and have bounded operator norms uniformly in

calibration. They are defined on arbitrary centers, not only on product posteriors or reachable representatives. In particular this reduction does not assume a factored decoder.

At most a constant times $Mu^k / \prod_b D_{b,\ell_b}$ of the minorized uniform rectangle can be covered by M balls of radius u . Taking $u = c_2(M^{-1} \prod_b D_{b,\ell_b})^{1/k}$ with c_2 small enough leaves at least half of that uniform mass uncovered. Mean squared prediction regret is therefore at least $c_3(M^{-1} \prod_b D_{b,\ell_b})^{2/k}$. Randomized assignment to centers cannot improve upon the closest center. Independent public coding randomness may be fixed first and averaged last, since it does not generate the acquisition history. Maximizing first over allocations and then over k proves the checkpoint lower bound by the factorial comparisons. Zero-volume allocations contribute zero and require no division. The unconditional optimum cannot exceed the maximum-history optimum. When no block is active, the true query vector is known in advance and all three regrets are zero.

One persistent index through the whole schedule. At every stage choose the preceding reachable joint codebook with at most M representatives. The initial codebook is the exact product prior. A representative is a tuple of block raw vectors stored in the read-only program; its index is the entire persistent state. When the next trial belongs to b , apply (24) only to its component, leave all other components unchanged, and quantize the updated tuple into the next joint codebook. The denominator in that block update is at least its report margin on a segment of posterior mixtures. Consequently the joint raw update is uniformly Lipschitz, including at all exponent collisions. An updated representative is reachable, so it is always inside the domain of the next covering quantizer. A previously unobserved block starts from its known exact prior, and a completed block is replaced by (1).

If e_s is the concatenated raw error, then

$$e_0 = 0, \quad e_{s+1} \leq L_s e_s + C_s \Theta_{s+1,\omega}(M, a)^{1/2}.$$

There are T stages and finitely many block updates, so iteration gives $e_s \leq C_T \Theta_\omega(M, a)^{1/2}$ for every s . This recurrence includes all errors from earlier joint re-encodings, including changes in inactive components. It never resets them by rereading a prefix. Lemma 8.1 yields the upper bound in (36).

Conversely, the index of any streaming filter is an M -message encoding at each checkpoint. All the checkpoint lower laws just proved are prefixes of the same uniform exploration experiment. Every filter therefore satisfies each such lower bound before an infimum is taken. Taking their maximum proves the streaming converse. Finally there are only finitely many count vectors and schedules for the fixed horizons; taking the largest upper and smallest positive lower constants makes all comparisons uniform in ω as asserted. $\square$

The tensor query algebra, the attainable posterior image, and the charged memory thus have different composition rules. The first is a tensor product, the second is a product embedded by a multilinear map, and the last is controlled by the maximum over allocations in (32). Only after that allocation is resolved can one take the maximum over causal checkpoints.

9. ADDITIVE BIT PROFILES AND THE EFFECT OF OBSERVATION ORDER

The preceding theorem determines not only the exponent at a fixed calibration but the cost of reallocating one memory through a schedule. For $0 < \varepsilon \leq 1$ define the nonnegative block function

$$(38) \quad b_b(n, \varepsilon, a^{(b)}) = \max_{0 \leq \ell \leq p_b(n, N_b - n)} \left\{ \frac{\ell}{2} \log_2 \frac{1}{\varepsilon} + \log_2 V_{b,\ell} \right\}.$$

Here the volumes are at the checkpoint $(n, N_b - n)$; the $\ell = 0$ term is zero, and a zero positive-index volume contributes $-\infty$. Thus $b_b(0, \varepsilon, a^{(b)}) = b_b(N_b, \varepsilon, a^{(b)}) = 0$. Set

$$L_b(\varepsilon, a^{(b)}) = \max_{0 \leq n \leq N_b} b_b(n, \varepsilon, a^{(b)}).$$

These are real-valued comparison profiles; they are not rounded integer budgets for a separately allocated machine.

Lemma 9.1 (Merging the attained scales). *At a checkpoint, sort the union of the first p_b Leja scales from all active blocks into $\delta_1 \geq \dots \geq \delta_{P_s} \geq 0$. Then*

$$(39) \quad A_{s,k} \asymp \prod_{j=1}^k \delta_j \quad (1 \leq k \leq P_s), \quad b_b(n_b, \varepsilon, a^{(b)}) = \sum_{j=1}^{p_b} \left[\log_2 \frac{d_{b,j}}{\sqrt{\varepsilon}} \right]_+ + O(1).$$

Zero scales contribute zero to the positive-part sum. All constants are uniform in calibration and ε . Thus the resolved axes under composition are the union of the attained block axes, not the products of all ambient tensor axes.

Proof. Within every block the scales are nonincreasing. With a tie rule respecting predecessors, the largest k members of their union form an initial segment in each block. They are therefore a feasible allocation in (32) after $V_{b,\ell}$ is replaced by $D_{b,\ell}$. No other selection of k scales has a larger product. The bounds $D_{b,\ell} \leq V_{b,\ell} \leq \ell! D_{b,\ell}$ give the first comparison, also when its two sides vanish. They change the maximum logarithmic expression for a block by a bounded amount. With $D_{b,\ell}$ that expression is the maximum of initial partial sums of the nonincreasing sequence $\log_2(d_{b,j}/\sqrt{\varepsilon})$. It equals the sum of its positive terms, with the empty partial sum allowed. Zero scales can be omitted before taking logarithms. This proves the second assertion. $\square$

Theorem 9.2 (Direct sum of persistent bit requirements). *Let $B_\omega^*(\varepsilon, a)$ be the smallest non-negative integer number of persistent bits with which one stage-compatible transducer has maximum-history, maximum-checkpoint regret at most ε in the product experiment. Then*

$$(40) \quad B_\omega^*(\varepsilon, a) = \max_{0 \leq s \leq T} \sum_{b=1}^B b_b(n_b(s), \varepsilon, a^{(b)}) + O(1).$$

The same formula holds for the maximum unconditional checkpoint criterion. The bounded remainder is uniform for all $0 < \varepsilon \leq 1$, all calibrations and all schedules of the fixed block horizons. In particular, arbitrary joint coding cannot decrease the right side by an unbounded number of bits.

Proof. For $M = 2^j$, solving all the inequalities $(A_{s,k}/M)^{2/k} \leq \varepsilon$ gives

$$j \geq F_\omega(\varepsilon, a) := \max_{s, 0 \leq k \leq P_s} \left\{ \frac{k}{2} \log_2 \frac{1}{\varepsilon} + \log_2 A_{s,k}(a) \right\}.$$

The $k = 0$ term is zero. By (32), the maximization at a fixed s is exactly

$$\begin{aligned} \max_{0 \leq \ell_b \leq p_b} \sum_b \left\{ \frac{\ell_b}{2} \log_2 \frac{1}{\varepsilon} + \log_2 V_{b,\ell_b} \right\} \\ = \sum_b b_b(n_b(s), \varepsilon, a^{(b)}). \end{aligned}$$

Terms with a zero factor are omitted on both sides, and each independent choice has its feasible zero index, so this separation is valid at exact collisions as well.

The constants c, C in Theorem 8.3 replace ε by a fixed constant multiple of ε in these inequalities. Every slope is at most $\max_s P_s/2$, a fixed number; such a replacement changes

F_ω by a bounded amount. Integer rounding costs at most one additional bit, and a state class with fewer than 2^j labels can be padded with unused labels. Applying both the upper and lower comparisons in (36) proves (40) for either criterion. This uses the joint converse of that theorem, not a restriction to product codes. $\square$

At a single checkpoint the equivalent expression is $\sum_{j=1}^{P_s} [\log_2(\delta_j/\sqrt{\varepsilon})]_+ + O(1)$. Consequently, up to bounded additive bits, composition introduces no new accuracy thresholds beyond those in the attained block scale lists. This concerns a fixed count vector; a change in which checkpoint is worst can of course introduce an additional transition in the maximum in (40). Neither statement identifies exact finite-budget thresholds or leading distortion constants.

Theorem 9.3 (Serial and overlapping schedules). *Let $\mathcal{W}$ be the finite set of schedules with the prescribed block horizons. Uniformly for $\varepsilon \in (0, 1]$ and $a \in \prod_b K_b$,*

$$(41) \quad \begin{aligned} \min_{\omega \in \mathcal{W}} B_\omega^*(\varepsilon, a) &= \max_b L_b(\varepsilon, a^{(b)}) + O(1), \\ \max_{\omega \in \mathcal{W}} B_\omega^*(\varepsilon, a) &= \sum_b L_b(\varepsilon, a^{(b)}) + O(1). \end{aligned}$$

Every serial schedule, completing each block before starting the next, attains the first formula. The same fixed serial schedule works for all calibrations and accuracies; it need not be selected anew. For the second formula one can choose a schedule that visits a count vector in which each block attains its own maximum. These assertions also hold for the unconditional bit criterion. For exact continuous memory, the corresponding identities hold without remainders:

$$(42) \quad \min_{\omega \in \mathcal{W}} D_\omega(a) = \max_b D_{A_b}(N_b), \quad \max_{\omega \in \mathcal{W}} D_\omega(a) = \sum_b D_{A_b}(N_b).$$

Proof. Every schedule traverses every integer local count $0 \leq n \leq N_b$ of every block. Since the functions in (38) are nonnegative, its maximum sum is at least $\max_b L_b$. In a serial schedule, at any checkpoint only one block can have a count strictly between its endpoints. The remaining blocks are either untouched or completed and have bit profile zero. Its maximum sum is therefore exactly $\max_b L_b$.

For every schedule the sum at every checkpoint is at most $\sum_b L_b$. Choose n_b^* maximizing the finite list for block b , and first execute n_b^* trials in each block, in any fixed block order. Then execute all remaining trials. This is an admitted schedule and visits the count vector $(n_b^*)_b$, where the sum is exactly $\sum_b L_b$. These are exact extrema for the real-valued profiles. Theorem 9.2 transfers them to integer bit budgets with a uniform bounded remainder. A maximizing schedule may depend on calibration and accuracy, unlike the serial minimizing schedule. The identical argument with the nonnegative integer functions $d_{A_b}(n, N_b - n)$ and Theorem 8.2 proves (42) without a remainder. $\square$

Corollary 9.4 (An order-dependent change of the sharp exponent). *Fix integers $B, h \geq 1$. In each of B independent blocks take $A_b = \{0, 1\}$, $N_b = 2h$, any fixed full-support prior, and the two positive cells*

$$k_0(t) = \frac{1}{2} + \frac{1}{4}t, \quad k_1(t) = \frac{1}{2} - \frac{1}{4}t.$$

Use a fixed admitted command cube in each block. A serial schedule has exact continuous peak dimension h and streaming regret $\asymp M^{-2/h}$. A schedule that first brings every block to count h has peak dimension Bh and streaming regret $\asymp M^{-2/(Bh)}$. Both use exactly $2Bh$ raw trials, one shared M -label memory, and the full tensor query protocol. The comparison holds for both regret criteria.

Proof. The cells sum to one, lie in $[1/4, 3/4]$, and have a coefficient matrix of rank two. Each block has $d_{A_b}(n, 2h - n) = \min(n, 2h - n)$ and peak h . Theorem 9.3 gives the two stated exact peaks, the latter being attained at the simultaneous count vector $(h, \dots, h)$. At this fixed calibration the positive determinant volumes are fixed positive numbers up to the number of distinct sites. Hence the largest nonzero index in (32) at a checkpoint is $\sum_b d_{A_b}(n_b, m_b)$. For $M \geq 1$ the corresponding profile has order $M^{-2/d_\omega(s,a)}$ when the dimension is positive. Taking its maximum and applying Theorem 8.3 proves the two regret laws. All trial counts and memory conventions are exactly those in the product experiment; no separate memories or uncharged copies of previous block histories are used. $\square$

The serial saving has a precise operational reason: once a block has no remaining trial, its posterior cannot affect a future-only query, and independence prevents it from changing the untouched blocks' priors. The conclusion does not authorize erasing a statistic that is explicitly requested by a retrospective payoff. It also makes no assertion about unknown calibration, dependent priors, adaptive schedule selection, or a horizon-uniform arithmetic implementation. Within the stated protocol, the upper and lower bounds nevertheless concern the unrestricted joint persistent state, rather than a prescribed allocation of separate block memories.

APPENDIX A. THE OBSERVATION ALGEBRA AND ASYMMETRIC PROFILES

For integral exponents define the standard graded algebra

$$\mathcal{R}_A = \mathbb{R}[zt^a : a \in A] \subset \mathbb{R}[z, t], \quad \deg(zt^a) = 1.$$

Its degree- m component has basis $\{z^m t^s : s \in mA\}$, so its Hilbert function is exactly $h_A(m)$. The general sumset–Hilbert–function connection is classical [2, 3]. Its role here is operational: $h_A(m) - 1$ is the number of independent future observable tests, and Theorem 4.3 proves when the attainable past can excite all of them. The algebra alone would not prove that transversality.

Proposition A.1 (Three-cell classification). *Let $A = \{0, a, b\}$ with integers $0 < a < b$, put $g = \gcd(a, b)$, $a' = a/g$ and $B = b/g$. Then*

$$(43) \quad h_A(m) = \binom{m+2}{2} - \mathbf{1}_{\{m \geq B\}} \binom{m-B+2}{2}.$$

The homogeneous relation is generated by $x_1^B - x_0^{B-a'} x_2^{a'}$ under the map $(x_0, x_1, x_2) \mapsto (z, zt^a, zt^b)$. In particular,

$$d_A(n, m) = \min \left\{ 2n, \binom{m+2}{2} - \mathbf{1}_{\{m \geq B\}} \binom{m-B+2}{2} - 1 \right\}.$$

For nonnegative real exponents $0 < a < b$ with a/b irrational, $h_A(m) = \binom{m+2}{2}$ for every m .

Proof. A monomial $x_0^k x_1^i x_2^j$ of degree m maps to $z^m t^{ai+bj}$. Two exponent triples of the same degree have the same image exactly when their difference is an integer multiple of $-(B - a'), B, -a'$: this follows by dividing $a(i - i') + b(j - j') = 0$ by g and using coprimality of a', B . After cancelling a common monomial, each corresponding binomial is a multiple of $x_1^B - x_0^{B-a'} x_2^{a'}$, by the difference-of-powers identity. The kernel is spanned by these binomials, because monomials with distinct images are linearly independent and each fiber's coefficient sum must vanish. Thus this one relation generates the kernel.

The degree- m polynomials in three variables have dimension $\binom{m+2}{2}$. Multiplication by the displayed nonzero degree- B relation is injective in the polynomial ring, so the degree- m part of the ideal has dimension zero for $m < B$ and $\binom{m-B+2}{2}$ otherwise. Subtraction proves (43). If

a/b is irrational, $a(i - i') + b(j - j') = 0$ forces both differences to vanish; all $\binom{m+2}{2}$ possible pairs are distinct. The operational formula now follows from Theorem 4.3. $\square$

Corollary A.2 (The sparse space that separates rank from future dimension). *For $A = \{0, 2, 5\}$, the first values of $h_A(m)$, starting at $m = 0$, are 1, 3, 6, 10, 15, 20, 25. At $n = 3, m = 2$ the exact information dimension is five, for every full-support prior. For a total budget of six the minimal causal dimensions, starting at $n = 0$, are*

$$0, 2, 4, 6, 5, 2, 0.$$

Thus the peak need not be the peak of a rank-based symmetric tent.

Proof. Apply (43) with $B = 5$ and then (2). In particular $2A = \{0, 2, 4, 5, 7, 10\}$, while the binomial witness at $n = 3$ has $B_{A,3} = \{0, 2, 5, 7, 10, 12, 15\}$. The mixed matrix has rank six and normalization leaves five. This proves the example uniformly over full-support priors without relying on a single large rational determinant. The uniform-prior three-cell example and its numerical witness first appeared in the source-pinned report [10]; the present corollary is an application of the general theorem, not a new attribution of that example. $\square$

There is also an exact finite computation for any integral A : start with $S_0 = \{0\}$, form $S_{m+1} = \{s + a : s \in S_m, a \in A\}$, and count distinct exponents. This handles all additive collisions, not just a nominal degree range. For real exponents the formula is equally exact as mathematics, but an effective implementation requires exact equality information about the supplied exponents. Neither fixed rank nor arbitrarily accurate floating-point exponents substitute for that information.

APPENDIX B. CONFLUENT OBSERVATION ALGEBRAS AND RESOLVED DIRECTIONS

The cardinality of a sumset is discontinuous when two of its elements coalesce. The corresponding experiment need not be discontinuous. We now identify the scales of the directions lost at such a collision. The parameter θ below indexes a *known calibration*; the unknown parameter of each experiment is still t .

B.1. Affine exponent families. Fix $0 = a_0 < a_1 < \dots < a_{r-1}$ and real numbers b_i , with $b_0 = 0$. On a sufficiently small compact interval $0 \leq \theta \leq \theta_0 \leq 1$ put

$$A_\theta = \{a_i + \theta b_i : 0 \leq i < r\}.$$

The order of these exponents is assumed unchanged. Take $I = [0, 1]$ and one fixed full-support prior μ . A fixed coefficient matrix specifies positive detector cells

$$(44) \quad k_j^\theta(t) = \sum_{i=0}^{r-1} c_{ji} t^{a_i + \theta b_i}, \quad \sum_j k_j^\theta = 1.$$

It has column rank r , and the cells have a common positive lower bound. The comparator cube is fixed. These hypotheses include any sufficiently small affine perturbation of a strictly positive calibration of full rank. No continuity of an unknown prior density is assumed.

For a fixed positive future length $m \geq 1$, let

$$\mathcal{B}_m = \left\{ \left(\sum_{\nu=1}^m a_{i_\nu}, \sum_{\nu=1}^m b_{i_\nu} \right) : 0 \leq i_\nu < r \right\}.$$

Repeated *pairs* are identified. Distinct pairs give distinct exponents $\lambda + \theta\gamma$ for $0 < \theta \leq \theta_0$; shrink θ_0 to exclude crossings if necessary. For each first coordinate λ , order its second coordinates as $\gamma_{\lambda,0} < \dots < \gamma_{\lambda,s_\lambda-1}$. Set $K_m = \sum_\lambda s_\lambda$. The cluster at $\lambda = 0$ consists only of $(0, 0)$.

Define the divided-difference functions

$$(45) \quad \psi_{\lambda,k}^\theta(t) = [\lambda + \theta\gamma_{\lambda,0}, \dots, \lambda + \theta\gamma_{\lambda,k}](x \mapsto t^x), \quad 0 \leq k < s_\lambda.$$

At $\theta = 0$ their continuous values are

$$(46) \quad \psi_{\lambda,k}^0(t) = t^\lambda (\log t)^k / k!.$$

For $\lambda > 0$ these functions have value zero at $t = 0$; the sole zero-exponent function is the constant one. The bracket in (45) is a divided difference in the exponent, not a new apparatus output. For a history h let $z_\theta(h)$ be its posterior expectations of these functions, omitting the constant coordinate.

List the orders k of the nonconstant functions in nondecreasing order:

$$0 = \nu_1 \leq \dots \leq \nu_d, \quad d = K_m - 1.$$

There are $|mA_0| - 1$ zero orders. The associated resolution function is

$$(47) \quad \Phi_\theta(M) = \max_{1 \leq \ell \leq d} \theta^{2(\nu_1 + \dots + \nu_\ell)/\ell} M^{-2/\ell}, \quad M \geq 1.$$

We set $0^0 = 1$ and $0^a = 0$ for $a > 0$ in this formula. In particular $\Phi_0(M) = M^{-2/(|mA_0|-1)}$.

Choose r attainable failure factors which form a basis in the formal monomials $t^{a_i + \theta b_i}$, with coefficients independent of θ . For example use $1/2$ and $1/2 + \delta t^{a_i + \theta b_i}$, $i > 0$, with δ sufficiently small. The fixed right inverse of the calibration matrix realizes these by fixed command labels. All ordered m -fold products form a common finite probe menu. Any other such fixed basis, and its full product menu, gives equivalent bounds below.

Theorem B.1 (Uniform resolved checkpoint law). *Suppose*

$$(48) \quad K_m - 1 \leq n(r - 1).$$

Supply independent uniform commands for the first n trials, then reveal an independent uniform query from the preceding product menu. Let $\bar{R}_{M;n,m}^\theta$ be the optimal unconditional squared-prediction regret of an arbitrary M -message checkpoint encoder. Let $R_{M;n,m}^{\theta,\max}$ be its minimax counterpart over admitted histories. There are constants $0 < c \leq C < \infty$, independent of M and θ , such that

$$(49) \quad c\Phi_\theta(M) \leq \bar{R}_{M;n,m}^\theta \leq R_{M;n,m}^{\theta,\max} \leq C\Phi_\theta(M), \quad 0 \leq \theta \leq \theta_0.$$

The encoders may be Borel and randomized as in Definition 3.1. At the common interior binomial command tuple the normalized map $h \mapsto z_\theta(h)$ is a submersion of rank d , including at $\theta = 0$. Its local inverse bounds and an unconditional minorization can be chosen uniformly in θ .

The upper bound in this theorem is a checkpoint bound. It does not assume that access to an exact prefix is free during repeated compression. Every streaming filter is a checkpoint encoder, so the lower bound applies to such filters as well. Section E proves a matching genuinely streaming bound across an entire singular family. Theorem D.1 below removes the full-future restriction by truncating the resolution flag, and Theorem D.2 proves the fixed-horizon causal version for the affine class. When $m = 0$ there is no nonconstant future query and the prediction regret is zero; formula (47) is not used in that case.

B.2. Confluent positivity and the physical prediction metric. We give the details of the analytic and geometric steps in Theorem B.1.

The strict mixed pairing used below is Lemma 5.2; its proof precedes the intrinsic theorem.

Lemma B.2 (Desingularization and observable scales). *The functions in (45), and the needed derivatives in θ , are uniformly bounded on $[0, 1] \times [0, \theta_0]$. The posterior vectors $z_\theta(h)$ therefore lie in a fixed box. For the common product probe menu let $p_\theta(h)$ be its probability vector. There is a fixed matrix G of full column rank such that*

$$(50) \quad p_\theta(h) - p_\theta(\tilde{h}) = G \operatorname{diag}(\theta^{\nu_i})(z_\theta(h) - z_\theta(\tilde{h})).$$

In particular the squared norm on the left is bounded above and below by positive constants times $\sum_i \theta^{2\nu_i} |z_i(h) - z_i(\tilde{h})|^2$, with constants independent of θ .

Proof. The divided difference of order k is the integral of $f^{(k)}(\sum_j w_j x_j)$ over the standard k -simplex, whose volume is $1/k!$. This identity follows by iterating the fundamental theorem of calculus, starting with $(f(y) - f(x))/(y - x) = \int_0^1 f'(x + s(y - x)) ds$. It also gives the repeated-node value. Here $f(x) = t^x$, so every integrand is $t^x (\log t)^k$. At a positive cluster the exponent is bounded below by some $a > 0$ uniformly on the fixed parameter interval. For every fixed j , $\sup_{0 < t \leq 1} t^a |\log t|^j < \infty$. The same domination applies after finitely many θ derivatives. There are no higher divided differences at zero. Posterior expectations of bounded functions are bounded independently of the history.

Newton's interpolation identity, proved by successive subtraction of the interpolating polynomials, is exact at each node:

$$(51) \quad t^{\lambda + \theta \gamma_{\lambda,j}} = \sum_{k=0}^j \theta^k \left(\prod_{v < k} (\gamma_{\lambda,j} - \gamma_{\lambda,v}) \right) \psi_{\lambda,k}^\theta(t).$$

The matrix multiplying $\operatorname{diag}(\theta^k)$ is fixed, triangular within each cluster, and has nonzero diagonal. The standard identities used here are classical divided-difference calculus [12]; the integral proof makes the uniformity at coalescence explicit.

The expansion matrix from the formal pair monomials in $\mathcal{B}_m$ to the probe products has full column rank. To see this, invert the fixed one-step basis change and multiply it m times. Every formal monomial product is a linear combination of the probe products. Identifying equal pairs is a surjective linear quotient, so the spanning statement survives that identification. Both this expansion matrix and the triangular matrix in (51) are independent of θ . Their product has full column rank. Subtracting two normalized posterior vectors removes its constant column and gives (50). The remaining columns are still independent. Their extreme singular values give the two constants. The equality extends to $\theta = 0$ by continuity; vanishing weights then remove precisely the unresolved jets. $\square$

Lemma B.3 (Uniform attainable patch with its evidence). *Under (48), there are $\rho, \beta > 0$, independent of θ , such that the subprobability distribution of z_θ on all-failure prefixes dominates β times uniform probability on a translate of $[-\rho, \rho]^d$.*

Proof. Use the binomial factors $(1 + c_i t^{D_\theta})/2$ of Lemma 4.2, with distinct small c_i and $D_\theta = a_{r-1} + \theta b_{r-1}$. Since the coefficient calibration is fixed, one common interior command tuple implements this family for every θ . Its unnormalized tangent has $n(r-1) + 1$ distinct monomials, also at zero. At positive θ , pairing with the K_m distinct future exponents has rank K_m by Lemma 4.1. The divided-difference basis change is invertible there. At zero, Lemma 5.2 gives the same rank directly, without passing a rank assertion through a singular limit. The product

itself lies in the tangent. The normalization map $v \mapsto v - pe_0v$ has kernel exactly $\text{span}\{p\}$ on this full image, as in (11). Thus the normalized jet map has rank d at every θ , including zero.

Here are uniform local details. Let q be the total number of command coordinates and g_* the common interior tuple. The matrices $J_\theta = D_g z_\theta(g_*)$ are continuous in θ and surjective. Their least row singular value has a positive minimum σ on the compact parameter interval. Choose an orthonormal basis E_θ of the row space and a complementary orthonormal basis F_θ of its kernel, and use coordinates $g = g_* + E_\theta u + F_\theta w$. The derivative at zero of

$$(u, w) \mapsto (z_\theta(g_* + E_\theta u + F_\theta w) - z_\theta(g_*), w)$$

is block diagonal with invertible upper block, least singular value at least σ , and lower block the identity. Its derivative norm and its second derivative have common finite bounds on a fixed neighborhood. This follows from Lemma B.2, finite product rules, and the prefix evidence lower bound κ^n .

Choose a ball inside the command cube on which the derivative differs from its central value by at most half its least singular value. This radius can be chosen uniformly. For a sufficiently small target ball, the map obtained by subtracting the nonlinear remainder and applying the inverse central derivative is a contraction of that command ball into itself. The contraction theorem gives a differentiable inverse. Hence a product of a d -cube and a $(q - d)$ -cube of common positive side length lies in its image, uniformly in θ . All inverse Jacobian determinants are bounded below by a positive constant: the forward derivative is uniformly bounded, so its determinant is uniformly bounded above. The orthogonal command-coordinate change has determinant of absolute value one. No continuity of the chosen orthonormal frames in θ is needed for these uniform bounds.

Uniform commands have density $(1 - 2\eta)^{-q}$. Their joint law with the all-failure word has density

$$(1 - 2\eta)^{-q} \mu \left(\prod_{i=1}^n F_{g_i} \right) \geq (1 - 2\eta)^{-q} \eta^n.$$

Change variables using the inverse just constructed and integrate over the complementary cube. This gives the asserted minorization, with $\beta > 0$ the product of this density bound, the inverse-Jacobian bound, and the two cube volumes. The convention for a zero-dimensional complement is volume one. This is a subprobability bound retaining the failure evidence, not conditioning away a rare prefix. $\square$

B.3. A scale-sensitive finite-state calculation.

Lemma B.4 (Rectangular quantization at every budget). *For $a_1 \geq \dots \geq a_d > 0$ set*

$$e_M(a) = \max_{1 \leq \ell \leq d} \left(\frac{a_1 \cdots a_\ell}{M} \right)^{1/\ell}.$$

A box with these side lengths has an at-most- M -cell rectangular partition of diameter at most $2\sqrt{d}e_M(a)$ in every cell. The optimal mean squared distortion of its uniform distribution using M arbitrary centers is at least $c_d e_M(a)^2$, with $c_d > 0$ depending only on d . Zero side lengths may be omitted.

Proof. Put $e = e_M(a)$ and take $N_i = \max\{1, \lfloor a_i/e \rfloor\}$ equal intervals in coordinate i . If ℓ coordinates satisfy $a_i \geq e$, then $\prod_i N_i \leq \prod_{i \leq \ell} a_i/e^\ell \leq M$. Every side length in a cell is at most $2e$: for $a_i/e \geq 1$ use $\lfloor a_i/e \rfloor \geq a_i/(2e)$, and otherwise use $a_i < e$. Choose an actual reachable representative in each nonempty cell when covering a reachable subset of the box. The diameter bound remains valid.

For the lower bound project onto the first ℓ coordinates. The projected uniform density is $(a_1 \cdots a_\ell)^{-1}$ on its box. The union of M balls of radius r has probability at most $Mv_\ell r^\ell / (a_1 \cdots a_\ell)$, where v_ℓ is the volume of the unit ball. Taking $r = (a_1 \cdots a_\ell / (2Mv_\ell))^{1/\ell}$ shows that mean squared distance is at least $r^2/2$. Projection cannot increase distance. Apply this for every ℓ and take the smallest of the finitely many positive dimensional constants. Centers need not lie in the box. $\square$

Proof of Theorem B.1. Let $D_\theta = \text{diag}(\theta^{\nu_i})$; this temporary matrix notation is unrelated to the peak dimension $D_A(N)$. By Lemma B.2, $D_\theta z_\theta(h)$ lies in a translated box of side lengths at most $2B\theta^{\nu_i}$ for a fixed B . Lemma B.4 partitions this box into at most M cells with squared diameter at most $C\Phi_\theta(M)$. Choose a reachable representative in each nonempty cell. At the checkpoint, encode the cell of the exact history and decode its representative's probe probabilities. These are probabilities in $[0, 1]$. Equation (50) gives the uniform upper regret bound. Finite cells with deterministic boundary conventions give Borel encoders.

For the lower bound first average private decoder coins conditional on the message and query. Squared loss cannot increase, leaving at most M centers in probe space. Orthogonally project these centers onto the affine space $p_* + \text{range}(G)$; projection cannot increase distance to a true prediction. Invert G on its range. Its lower singular bound reduces the problem to distance from at most M arbitrary centers in the scaled z coordinates. The law there dominates β times the image of the cube from Lemma B.3; its side lengths are $2\rho\theta^{\nu_i}$. The lower half of Lemma B.4 gives $c\Phi_\theta(M)$ after division by the fixed number of queries. Encoder randomization cannot beat distance to the closest center at a fixed history. Fresh public randomness can be fixed and then averaged because it is independent of the entire prefix. At zero, omit zero-weight coordinates in this argument. The same constants, reduced if necessary over the finitely many possible dimensions, work throughout the compact parameter interval. This proves both criteria and the claimed submersion statement. $\square$

Thus the relevant finite-resolution data are the multiplicities of additive collisions and their confluent orders, not only the cardinality of the nearby sumset. The result is a uniform comparison up to constants. It neither asserts a new general quantization principle nor identifies a universal leading distortion constant.

APPENDIX C. FIXED-CALIBRATION STREAMING AND CONTROL

Fix $N \geq 2$ and abbreviate $d_n = d_A(n, N - n)$ and $D_* = D_A(N) > 0$. The exact state from Theorem 4.4 will be denoted s_n . A clock n , calibration, prior moments and a precomputed finite program are read-only data. The charged resource is the persistent state between successive input symbols.

Lemma C.1 (Uniform finite-horizon transition bounds). *On the compact exact state sets there are constants $L_n < \infty$ such that the update $T_n(s, g, x)$ satisfies*

$$\|T_n(s, g, x) - T_n(\tilde{s}, g, x)\| \leq L_n \|s - \tilde{s}\|$$

for every admitted g, x and $s, \tilde{s} \in \mathcal{S}_n$. Every fixed finite probe prediction vector is likewise Lipschitz in the state, with a finite constant independent of the history.

Proof. In factor coordinates an update before changeover appends the new factor $f/\mu f$. The old factor coordinates are unchanged. All unnormalized report probabilities lie in $[\kappa, 1]$; consequently the normalized factors lie pointwise in $[\kappa, 1/\kappa]$. Their coordinates are bounded, since they are continuous images of a compact cube and finite alphabet. At changeover each needed moment is a ratio of polynomial expressions in these coordinates and the fixed prior moments, with

denominator the integral of a product of normalized factors, bounded below by κ^{n+1} . The same lower bound holds on line segments between factor tuples, because each individual interpolated normalized factor is still at least κ . Finite differentiation and the mean value theorem give a uniform derivative bound there.

In moment coordinates the update is (12). Its denominator is at least κ at every state; on a line segment between two such moment vectors the corresponding posterior mixture has the same lower bound. Numerators, coefficients and moment coordinates range over compact sets. The quotient rule therefore gives a uniform bound. Probe predictions are linear in the relevant moment coordinates, or normalized polynomial expressions in the factor coordinates with the same positive denominator control. The argument gives their claimed Lipschitz constants. All constants use only finite algebra and the supplied moments; they may grow with N or deteriorate with κ . $\square$

Theorem C.2 (Finite-state streaming upper bound). *For every $M \geq 1$ there is a deterministic M -state filter, updated after every report, whose prediction regret at every checkpoint and every admitted command-report history is at most*

$$(52) \quad C_N M^{-2/D_*},$$

where $C_N < \infty$ depends on the fixed experiment, prior, probe lists and total horizon, but not on M or the input history. The representatives and read-only transition functions are supplied existentially from the exact model; this is not an effective synthesis theorem from finite-precision data.

Proof. Enclose $\mathcal{S}_n \subset \mathbb{R}^{d_n}$ in $[-B_n, B_n]^{d_n}$. For $d_n > 0$ partition each coordinate interval into $k_n = \lfloor M^{1/d_n} \rfloor$ intervals. In each nonempty cell choose one representative in $\mathcal{S}_n$. There are at most $k_n^{d_n} \leq M$ representatives, and assigning a state to its cell representative has error at most

$$\delta_n \leq 2B_n \sqrt{d_n}/k_n \leq 4B_n \sqrt{d_n} M^{-1/d_n} \leq CM^{-1/D_*}.$$

Here C is the maximum of the finite prefactors. Empty cells need no representative. For $d_n = 0$ use the unique state. Deterministic boundary and tie conventions give Borel quantizers Q_n .

The filter stores only the index of a representative $\hat{s}_n$. Its transition is exactly

$$(53) \quad \hat{s}_{n+1} = Q_{n+1}(T_n(\hat{s}_n, g_{n+1}, x_{n+1})).$$

Since each representative is reachable and every report has positive probability, its exact image lies in $\mathcal{S}_{n+1}$. Thus the quantizer is always applied inside its domain and no infeasible moment update is used. The finite representative choices and the functions in (53) are read-only program data. They may be obtained by selecting points in nonempty cells of the compact state sets; no efficient global synthesis or computability of unsupplied real prior moments is asserted.

Let $e_n = \|s_n - \hat{s}_n\|$. On the actual command-report stream, Lemma C.1 gives

$$e_0 = 0, \quad e_{n+1} \leq L_n e_n + CM^{-1/D_*}.$$

Set $b_0 = 0$ and $b_{n+1} = L_n b_n + C$. Induction proves $e_n \leq b_n M^{-1/D_*}$. This explicitly accumulates the errors from *all* previous lossy updates. At a query use the probe predictions of $\hat{s}_n$, which are actual probabilities. If the vector prediction map has Lipschitz constant K_n , (7) bounds regret by $(K_n^2/s_n^q)b_n^2 M^{-2/D_*}$. The maximum of these finitely many constants proves (52). No exact prefix was kept for the changeover: the factor-to-moment computation uses the representative factors already encoded by i_n . $\square$

Theorem C.3 (Matching exponent in one fixed streaming experiment). *Choose n_* attaining $D_* = d_A(n_*, N - n_*)$. For the first n_* trials supply independent uniform commands from $\mathcal{G}$, independent of the parameter, cartridges and filtering coins. The realized command is read once*

by the filter. At n_* reveal the independent probe query and execute its remaining trials. This is one fixed experiment, known before the filter is chosen. Its optimal unconditional regret over M -state filters satisfies

$$(54) \quad 0 < c_N M^{-2/D_*} \leq \overline{\mathcal{R}}_{M,N}^{\text{stream}} \leq C_N M^{-2/D_*}.$$

The same power is sharp for the minimax streaming criterion taking the maximum conditional regret over checkpoints and input histories. Fresh private update and decoder randomization does not defeat the lower bound; a fresh public coding seed independent of the complete prefix may also be allowed. Thus for fixed N , the required persistent bits for this prediction task satisfy $B_*(\varepsilon) = \frac{D_*}{2} \log_2(1/\varepsilon) + O_N(1)$.

Proof. Write $n = n_*$, $d = D_*$, and $s = s_n^q$. Theorem 4.3 gives an interior command tuple g_0 where the probe map $p(g)$ on all-failure prefixes has rank d . Select d independent probe coordinates, and let E denote their coordinate projection, of operator norm one. Choose complementary command coordinates w so that $g \mapsto (Ep(g), w(g))$ has an invertible derivative at g_0 . Let Φ be its local inverse. There is a compact product $B_\rho^d \times W$ inside its domain, centered at the projected prediction point, such that Φ stays in the command interior and $|\det D\Phi| \geq j_0 > 0$. The complementary box W has positive volume, with unit-volume convention in zero dimensions.

If $k = Kn$ is the number of real command entries, their density is $(1 - 2\eta)^{-k}$. The joint density of commands and an all-failure prefix is this density times $Z(g) = \mu \prod_i F_{g_i} \geq \eta^n$. Change variables with Φ and integrate over W . The subprobability law of the projected predictions on this prefix dominates β times uniform probability on B_ρ^d , with

$$\beta = \frac{\eta^n j_0 |W| |B_\rho^d|}{(1 - 2\eta)^k} > 0.$$

In particular the lower bound retains the probability of the failure prefix. It does not assign an atom to an exact real command or assume a flat ball in the entire curved prediction image.

Every streaming filter is in particular an M -message encoder at this checkpoint. Given any message, average decoder coins conditional on each query; squared loss can only decrease. This gives at most M prediction vectors. Randomizing the encoder cannot beat the closest of these vectors at a fixed history. Projecting their centers by E cannot increase squared distance. For a uniform point X in a d -ball of radius ρ and any M centers, the union-of-balls estimate gives

$$\Pr\{\min_i \|X - z_i\| \leq r\} \leq \min\{1, M(r/\rho)^d\}.$$

Integrating its complementary tail yields

$$\mathbb{E} \min_i \|X - z_i\|^2 \geq \int_0^{\rho^2 M^{-2/d}} (1 - Mt^{d/2}/\rho^d) dt = \frac{d}{d+2} \rho^2 M^{-2/d}.$$

Multiply by β/s to obtain $c_N > 0$. Fresh public randomness can be fixed and then averaged because it is independent of the prefix. Private online coins may equivalently be drawn on an independent tape for this proof; fixing them makes a deterministic transducer and the same bound holds uniformly. That proof device supplies no tape to the implemented filter. The prescribed exploration is independent of that tape, a condition that is essential to this particular argument. An old command-generation seed is not fresh coding randomness and remains charged.

Theorem C.2 gives the upper bound in the same experiment. The minimax conditional criterion is at least its unconditional regret, while the universal upper filter works at all checkpoints. Finally put $M = 2^B$ and solve the two inequalities. The Euclidean covering

mechanism is classical [4]; the identified sparse dimension and the actual repeated-update construction are the experiment-specific inputs. $\square$

Theorem C.4 (A separate finite-state control bound). *For the same finite detector and horizon, suppose stage rewards $r_n(g, x) \in [0, 1]$ are continuous in g for each report, and the objective is their expected sum. There is an M -state feedback controller whose loss relative to the full-history optimum is at most*

$$(55) \quad C'_N M^{-1/D_*}.$$

No matching lower power for arbitrary control tasks is asserted. An additional finite payoff automaton must be counted as task state, multiplying the number of states if it is included.

Proof. Exact dynamic programming on $\mathcal{S}_n$ is valid by Proposition 2.3. With terminal value zero, its recursion is

$$V_n(s) = \max_{g \in \mathcal{G}} \sum_{x \in \mathcal{X}} p_x(s, g) \{r_{n+1}(g, x) + V_{n+1}(T_n(s, g, x))\}.$$

Here p_x is a linear moment functional or a normalized factor expression. The positive denominator bounds, compact action set and continuous rewards make the displayed objective continuous, and the maximum is attained. By backward induction it is uniformly Lipschitz in s , since finite sums, products of bounded Lipschitz functions, and the updates in Lemma C.1 preserve that property. The maximum of functions with a common Lipschitz constant is again Lipschitz. Denote by H_n a uniform Lipschitz constant of the action objective, before its maximization.

At each representative choose an exact maximizing action and store that action in the finite output table. At the true state s_n , the loss from using the action optimal at $\hat{s}_n$ is at most $2H_n e_n$. Update the representative by (53). The true exact state is computed only for analysis along the controller's actual observations; the implemented controller still stores only its index. Bellman telescoping and conditional expectation give total loss at most $2 \sum_{n < N} H_n e_n \leq 2 \sum_{n < N} H_n b_n M^{-1/D_*}$. This proves (55). It uses no continuity of the optimizing action as a function of state, and it does not differentiate an optimizer. Its approximation principle is related to [8]; it is not claimed as a new general dynamic-programming principle. $\square$

C.1. Horizon and calibration remain part of the constants. If every factor in a probe menu satisfies $0 < F \leq v < 1$, then $H_{n,j} \leq v^{N-n}$ and every posterior prediction satisfies the same bound. The one-state decoder that always announces zero therefore has

$$(56) \quad \mathcal{R} \leq v^{2(N-n)}.$$

For the shared physical menu $F = c + \delta k_j$ one may take $v = c + \delta$. Thus a checkpoint implementation may choose, in advance, between the streaming filter and the one-state decoder and obtain

$$\mathcal{R} \leq \min\{C_N M^{-2/D_*}, (c + \delta)^{2(N-n)}\}.$$

At $c = 1/2$, $\delta = 1/8$, $N - n = 20$, the second bound is $(5/8)^{40} < 10^{-8}$. This does not contradict a fixed- N high-resolution exponent. Neither the new sparse rank theorem nor the streaming construction turns it into a horizon-uniform useful-memory law. We have not rescaled the loss or discarded rare histories to avoid this bound.

The upper constants involve state bounds, transition Lipschitz constants and repeated error propagation. The lower constants involve an attainable Jacobian, a chart radius, evidence and the prior pairing. They are not uniform over vanishing detector amplitudes, all full-support priors or increasing horizons. With a finite fixed command alphabet and finite N there are finitely many histories and an exact finite code eventually suffices; the matching asymptotic lower bound above expressly uses the fixed continuous-command exploration protocol.

APPENDIX D. ATTAINABLE RESOLUTION FLAGS AND UNIFORM CAUSAL MEMORY

The full-future hypothesis in Section B makes every nonconstant future jet attainable. It is not needed for a resolution law, but it cannot simply be omitted from the formula there. When the past is shorter, the relevant list is an initial part of the future resolution flag. The lower bound requires transversality to this flag; the upper bound requires an estimate on the whole, generally curved, attainable image. We establish both and then prove their compatibility with every successive report update.

D.1. Statement of the attainable law. Retain the affine calibration and positivity assumptions of Section B, but do not impose (48). Fix $n, m \geq 1$. There are K_m formal sum pairs in $\mathcal{B}_m$; the zero pair is the constant, and the other $q_m = K_m - 1$ divided-difference coordinates have orders

$$\nu_{m,1} \leq \cdots \leq \nu_{m,q_m}.$$

Order equal orders by increasing limiting exponent, once and for all. Thus every initial part of the list includes, within each cluster, all jets preceding any jet that it contains. Set

$$(57) \quad p_{n,m} = \min\{n(r-1), q_m\}, \quad S_{m,\ell} = \sum_{i=1}^{\ell} \nu_{m,i}, \quad \Phi_{n,m}^{\text{att}}(M, \theta) = \max_{1 \leq \ell \leq p_{n,m}} \theta^{2S_{m,\ell}/\ell} M^{-2/\ell}.$$

Here $0^0 = 1$ and $0^s = 0$ for $s > 0$. If $n = 0$ or $m = 0$, define $\Phi_{n,m}^{\text{att}} = 0$ separately. No division by a zero dimension is then involved. Throughout, constants may depend on the fixed experiment, prior, horizon and calibration interval, but not on M or θ .

Theorem D.1 (Attainable confluent checkpoint law). *For every integer $M \geq 1$ and every calibration in the compact interval of Section B, the unconditional optimal M -message checkpoint regret under independent uniform commands and the minimax checkpoint regret both have order $\Phi_{n,m}^{\text{att}}(M, \theta)$, with positive uniform comparison constants. The upper code has at most M reachable representatives and works on every admitted history. The lower bound permits the independent encoder and decoder randomization specified in Definition 3.1.*

In particular, past limitation truncates the ordered ambient list at $p_{n,m}$; it does not introduce the remaining ambient orders into the rate. At zero the exponent is $2/\min\{n(r-1), |mA_0| - 1\}$, and at each fixed positive calibration it is $2/p_{n,m}$.

For the sequential assertion fix $N \geq 2$ and choose the compact calibration interval so that distinct formal pairs in $\mathcal{B}_m$, $1 \leq m < N$, do not meet at a positive calibration. Such an interval always exists after decreasing its positive endpoint: there are finitely many pairs and their exponent differences are affine. Put

$$(58) \quad \Omega_N(M, \theta) = \max_{1 \leq n < N} \Phi_{n, N-n}^{\text{att}}(M, \theta).$$

Use the same independent uniform command law at every acquisition time. For each selected checkpoint reveal its independent product query only after the index has been formed. As in Section C, only one checkpoint and its remaining query trials are executed in a run. Let $\mathfrak{R}_{M,N}^{\text{av}}$ be the infimum, over one streaming filter, of the maximum of its unconditional checkpoint regrets. Let $\mathfrak{R}_{M,N}^{\text{max}}$ be the corresponding infimum of the maximum conditional regret over checkpoints and histories.

Theorem D.2 (Uniform causal realization of the resolution flag). *In this fixed-horizon experiment there are $0 < c \leq C < \infty$ such that*

$$(59) \quad c\Omega_N(M, \theta) \leq \mathfrak{R}_{M,N}^{\text{av}}(\theta) \leq \mathfrak{R}_{M,N}^{\text{max}}(\theta) \leq C\Omega_N(M, \theta)$$

for every $M \geq 1$ and every admitted θ , including zero. For each known θ and each M , there exists a single deterministic M -state transducer attaining the upper bound after every report. Its codebooks and transitions may depend on θ and M ; the comparison constants do not. No joint-in-calibration measurable or computable selection of codebooks is asserted. It stores only a representative index; its transitions use physical moments, never division by a collision parameter. Read-only prior moments at the formal exponents in $\mathcal{B}_N$ suffice. The lower bound holds in the stated common exploration experiment, including for the independent randomized filters of Definition 3.1.

These are statements about finite persistent memory in the exact-real program model already defined, not about finite-precision synthesis. In particular the number of real coordinates used to describe a representative is not the persistent state count: the latter is the number of representative labels, at most M .

D.2. Format and the attained flag. The covering estimate needed for the whole image is Lemma 6.1, proved before the intrinsic classification.

Lemma D.3 (The whole attainable image has bounded format). *Let $z_\theta(h) \in \mathbb{R}^{q_m}$ be the nonconstant divided-difference moment vector, in the ordered coordinates of (57), and put*

$$D_{m,\theta} = \text{diag}(\theta^{\nu_{m,i}}), \quad S_{n,m}^\theta = \{D_{m,\theta} z_\theta(h) : |h| = n\}.$$

The set $S_{n,m}^\theta$ is compact, has semialgebraic format bounded independently of θ , and has dimension at most $p_{n,m}$. It is contained in a rectangle with sides $[-B\theta^{\nu_{m,i}}, B\theta^{\nu_{m,i}}]$, with B uniform. At zero all positive-order coordinates are zero.

Proof. Fix the finite report word $x_1, \dots, x_n$ and write u for its command entries. Every report factor is affine in its current command. Thus the evidence $Z(u) = \mu \prod_{i=1}^n f_{x_i, g_i}^\theta$ and each numerator

$$N_j(u) = \mu \left(\psi_j^\theta \prod_{i=1}^n f_{x_i, g_i}^\theta \right)$$

are polynomials of degree at most n in u . Their coefficients are real prior integrals, well defined also at zero. The moment image is specified by the existence of u in its fixed cube with $Z(u)y_j = \theta^{\nu_{m,j}} N_j(u)$ for every j . Moreover $Z(u) \geq \kappa^n > 0$. This is a semialgebraic formula of fixed format, uniformly in its real coefficients. Finite union over the report words preserves a format bound, and continuity on compact command cubes proves compactness. There is no requirement that these real coefficients, as functions of θ , be semialgebraic.

For the dimension bound, divide each report factor by its prior integral. Its coefficient vector lies in the affine hyperplane $\mu f = 1$ of the r -dimensional one-step space. The constant monomial allows its constant coefficient to be eliminated, leaving at most $r - 1$ coordinates. The product posterior and all displayed moments are rational functions of these $n(r - 1)$ coordinates. A rational map cannot increase semialgebraic dimension. Multiplication by $D_{m,\theta}$ cannot increase it either; the ambient dimension is q_m . Finally the uniform bounds for ψ_j^θ in Lemma B.2 give $|z_j| \leq B$ before any command integration: they are expectations under probability measures. This proves the rectangle assertion without an inverse power of the prefix evidence. $\square$

D.3. Transversality to every initial resolution flag.

Lemma D.4 (Attainment of the truncated flag). *At one common interior binomial command tuple, projection of $z_\theta(h)$ onto its first $p_{n,m}$ coordinates has full row rank for every calibration, including zero. Under independent uniform commands, the all-failure subprobability distribution of this projection dominates β times uniform probability on a translated cube of side 2ρ , where $\beta, \rho > 0$ are independent of the calibration.*

Proof. Choose distinct sufficiently small $c_1, \dots, c_n$ and realize the factors $(1 + c_i t^{D_\theta})/2$ by the fixed right inverse of the calibration matrix, as in Lemma 2.1. Their commands stay in a common interior neighborhood. Lemma 4.2 identifies their unnormalized tangent with a monomial space of dimension $n(r-1) + 1$, for every calibration, including zero.

Adjoin the constant to the selected $p = p_{n,m}$ future tests. Within each limiting exponent cluster, the selected tests are exactly an initial block of its divided differences. At a positive calibration this block is an invertible linear transformation of the monomials at its first nodes. Their union consists of distinct monomials. The mixed pairing with the binomial tangent therefore has column rank $p + 1$, by Lemma 4.1. At zero these same blocks become complete initial logarithmic-jet blocks. Lemma 5.2 gives column rank $p + 1$ directly at zero. This is not an inference of rank at zero from rank on a punctured interval.

Let L be this pairing, let F be the product at the witness, and write $v = LF/\mu F$, with constant coordinate $e_0 v = 1$. The normalized differential has image

$$(\mu F)^{-1} \{ LQ - v e_0 LQ : Q \text{ in the product tangent} \}.$$

Since L is onto $\mathbb{R}^{p+1}$, the bracket is precisely the p -dimensional hyperplane $e_0 = 0$. Deleting the constant proves the asserted row rank. The remaining, unselected future coordinates need not be independent and are not included in this submersion assertion.

For clarity we give the uniformity and probability argument. On the compact calibration interval, the least row singular value of the selected derivative has a positive minimum, by the rank just proved and continuity of all divided-difference derivatives. The second command derivatives have a common bound, since the factors stay positive and the test functions are uniformly bounded. Complete the row coordinates with orthonormal kernel coordinates. On a fixed small command ball the resulting square derivative differs from its value at the center by at most half its uniformly positive least singular value. The contraction proof of the inverse theorem gives a common product box in the selected output and complementary coordinates. Shrink it so its inverse stays in the command interior. Uniform forward derivative bounds give a positive lower bound for the absolute inverse Jacobian. These estimates use only orthogonality of the chosen frames; a continuous choice of frames is unnecessary.

If k is the number of command entries, their joint density is $(1 - 2\eta)^{-k}$. The density of commands together with the all-failure word is this density multiplied by its evidence, at least η^n . Change variables using the local inverse and integrate over the common complementary box. The resulting measure dominates a fixed positive multiple of Lebesgue measure on the selected p -cube. Absorb the cube volume into β . This proves the stated subprobability minorization; it does not condition away the probability of the failure word. A full-support prior without a density is permitted throughout, since the continuous density used here is that of the commands. $\square$

Proof of Theorem D.1. The physical prediction identity of Lemma B.2 reads, after the fixed coordinate permutation,

$$(60) \quad p_\theta(h) - p_\theta(\tilde{h}) = G_m D_{m,\theta}(z_\theta(h) - z_\theta(\tilde{h})),$$

where G_m has full column rank and is independent of θ . Apply Lemmas 6.1 and D.3 to the complete attainable image. Since $0 \leq \theta \leq 1$, its ordered side lengths are $B\theta^{\nu_{m,i}}$. Equation (15) supplies at most M reachable representatives with squared covering radius at most $C\Phi_{n,m}^{\text{att}}$. The fixed norm of G_m and the finite query average convert this into the required uniform prediction upper bound. This is a cover of all report strata, not just the local patch used in the lower bound.

For that lower bound, first average decoder randomization for each message and query. The outputs form at most M centers. Orthogonally project them to the affine span $p_* + \text{im } G_m$ and apply the fixed left inverse of G_m . Distances can decrease only by a fixed factor. Project further onto the first $p_{n,m}$ coordinates. Lemma D.4 then gives a subprobability multiple of uniform measure on a translated rectangle of side lengths $2\rho\theta^{\nu_{m,i}}$, $i \leq p_{n,m}$. The lower half of Lemma B.4, projecting onto each first ℓ axes, gives $c\Phi_{n,m}^{\text{att}}$. Centers may be outside the image; randomized assignment cannot improve on the nearest center. Fixing and averaging an independent public coding seed gives the same bound. At zero, omit the zero-width axes. There are $\min\{n(r-1), |mA_0| - 1\}$ surviving axes, which also proves the stated limiting exponent. For positive calibration the last power $M^{-2/p_{n,m}}$ dominates as $M \rightarrow \infty$. The unconditional lower bound is also a minimax lower bound, completing the proof. $\square$

D.4. Physical moments and successive updates. For $b = (\lambda, \gamma) \in \mathcal{B}_m$, write

$$v_b(h) = \mu(P_h t^{\lambda+\theta\gamma}), \quad v_{(0,0)} = 1.$$

We keep the formal labels even if the functions coincide at zero. Let $v^{(m)}$ denote the nonconstant vector. Newton interpolation within each cluster, as in Lemma B.2, gives a fixed invertible matrix T_m with

$$(61) \quad v^{(m)} = T_m D_{m,\theta} z_\theta.$$

The constant cluster has no other member, so no deleted constant mixes with this equation. Both T_m and its inverse have fixed finite norms. At zero the equation still holds with the appropriate repeated raw moments. It is the analysis of this equation, not inversion of its vanishing diagonal, that identifies the metric scales.

Lemma D.5 (Uniform updates in the physical metric). *The map from reachable raw moments at remaining length m to those at length $m-1$, after a command and a report, is uniformly Lipschitz in the raw Euclidean metric, independently of calibration. Every reachable representative maps to a reachable next vector.*

Proof. Write $e_i = (a_i, b_i)$ and the current report factor as $f = \sum_i f_i(g, x) t^{a_i + \theta b_i}$. Its coefficients are uniformly bounded, and $f \geq \kappa > 0$. Addition of formal pairs gives $b + e_i \in \mathcal{B}_m$ whenever $b \in \mathcal{B}_{m-1}$. Consequently the exact update is

$$(62) \quad v'_b = \frac{\sum_i f_i(g, x) v_{b+e_i}}{\sum_i f_i(g, x) v_{e_i}}, \quad b \in \mathcal{B}_{m-1}.$$

Every denominator is the posterior expectation of f , hence at least κ . A line segment between reachable raw vectors represents a mixture of their posterior probability measures, so the same bound holds along the segment. All numerators, coefficients and coordinates have uniform bounds. The quotient rule and the mean value theorem prove the Lipschitz assertion. No exponent separation occurs in a denominator. The formula is valid also when formal labels have equal values. Finally a history realizing the representative, followed by the specified input, realizes its updated vector; positivity admits every report. $\square$

Proof of Theorem D.2. For each $1 \leq n < N$, Lemmas 6.1 and D.3, followed by the fixed matrix T_{N-n} , give a raw-moment quantizer Q_n with at most M reachable representatives and radius

$$\delta_n \leq C_n \sqrt{\Phi_{n,N-n}^{\text{att}}(M, \theta)} \leq C_0 \sqrt{\Omega_N(M, \theta)}.$$

The maximum C_0 is over finitely many stages. At time zero the prior moment vector is one fixed state; at time N the remaining state is constant. Initialize at the prior vector and implement

$$\hat{v}_{n+1} = Q_{n+1}(U_n(\hat{v}_n, g_{n+1}, x_{n+1})),$$

where U_n is (62). The transition reads the old representative index and the current input only. Reachability makes its argument an element of the next quantizer's domain. The finite codebooks and these real-valued functions are read-only program data.

Let $e_n = \|v_n - \hat{v}_n\|$. With $e_0 = 0$, Lemma D.5 yields

$$e_{n+1} \leq L_n e_n + C_0 \sqrt{\Omega_N(M, \theta)}.$$

Set $b_0 = 0$ and $b_{n+1} = L_n b_n + C_0$. Induction gives $e_n \leq b_n \sqrt{\Omega_N(M, \theta)}$ after every update. Each query prediction is a linear form with fixed coefficients in the raw formal moments. Its Lipschitz constant is therefore uniform, and (7) proves the last inequality in (59). There is no exact-prefix recomputation: all intermediate vectors in this implementation are functions of a finite representative label and the current command-report pair.

For every streaming filter, its index at checkpoint n is an M -message encoder of the prefix. Theorem D.1 therefore bounds that checkpoint's unconditional regret below by $c_n \Phi_{n, N-n}^{\text{att}}$. The commands in all these assertions are prefixes of the same independent uniform acquisition rule. Taking the maximum over n , and then the infimum over filters, gives the first inequality with $c = \min_n c_n > 0$. Independent filtering coins can be fixed for this argument because they do not generate the acquisition commands; averaging restores the randomized statement. Finally a maximum conditional risk dominates its unconditional average, which proves the middle inequality. $\square$

Corollary D.6 (A uniform bit profile). *Let $B_N(\varepsilon, \theta)$ be the least number of persistent bits needed to make the minimax streaming regret at most ε . For $0 < \varepsilon < 1$, uniformly in the admitted calibration,*

$$(63) \quad B_N(\varepsilon, \theta) = \max \left\{ 0, \max_{\substack{1 \leq n < N \\ 1 \leq \ell \leq p_{n, N-n}}} \left(\frac{\ell}{2} \log_2 \frac{1}{\varepsilon} - S_{N-n, \ell} \log_2 \frac{1}{\theta} \right) \right\} + O(1).$$

At zero retain only terms with $S_{N-n, \ell} = 0$. The same formula holds for the maximum unconditional criterion.

Proof. For positive θ , the condition $\Omega_N(M, \theta) \leq \varepsilon$ is equivalent to all inequalities $M \geq \theta^{S_{N-n, \ell}} \varepsilon^{-\ell/2}$. Taking the largest right-hand side, adjoining $M \geq 1$, and then taking base-two logarithms gives the expression. The two fixed comparison constants in (59) change it by a bounded amount, since the possible ℓ are bounded by the fixed horizon. Rounding to a power of two changes it by at most one further bit. The zero-calibration statement follows by omitting vanishing terms. $\square$

D.5. A past-limited peak with two weak directions. The next consequence answers the concrete seven-trial boundary in the source-pinned v7 technical note [19]. It is a consequence of the attainable-image argument, rather than an extrapolation of the full-future formula.

Corollary D.7 (Seven-trial resolved memory). *For the cells (65), their fixed command cube and the product probes (66), take total horizon seven. For the two streaming criteria above, uniformly for $0 \leq \theta \leq 1/2$ and $M \geq 1$,*

$$(64) \quad \mathfrak{R}_{M,7}(\theta) \asymp \max\{M^{-1/3}, \theta^{1/2} M^{-1/4}\}.$$

The lower bound already holds at checkpoint four under the same uniform exploration. The crossover budget has order θ^{-6} and the crossover regret has order θ^2 .

Proof. Put $D = 2 + \theta$. For positive $\theta \leq 1/2$, the ten distinct three-fold exponents are

$$0, 1, 2, D, 3, D+1, D+2, 2D, 2D+1, 3D.$$

At zero their limiting clusters have multiplicities 1, 1, 2, 2, 2, 1, 1 at exponents 0, 1, 2, 3, 4, 5, 6. Thus the nonconstant order list has six zeros and three ones. At $n = 4, m = 3$, however, $p_{4,3} = 8$, and the selected list has six zeros and only two ones. Theorem D.1 gives

$$\Phi_{4,3}^{\text{att}} = \max\{M^{-1/3}, \theta^{2/7} M^{-2/7}, \theta^{1/2} M^{-1/4}\} = \max\{M^{-1/3}, \theta^{1/2} M^{-1/4}\}.$$

The middle term is the product of the first term to power 3/7 and the last to power 4/7, so cannot exceed their maximum. The minorized eight-dimensional flag uses, at zero, the tests

$$1, t, t^2, t^2 \log t, t^3, t^3 \log t, t^4, t^5, t^6$$

paired with the binomial tangent $\mathcal{P}_8$. Strict confluent positivity proves its rank for every full-support prior. The third ambient weak jet is not counted as an additional attainable dimension.

For completeness the extension of the upper bound to the whole stated calibration interval does not require excluding positive collisions in longer, unused sumsets. At stages $n = 1, 2, 3$, the raw reachable images have dimension at most $2n \leq 6$ and bounded format, and all raw coordinates lie in $[0, 1]$. Lemma 6.1 gives squared covering errors at most $CM^{-1/3}$ uniformly, whether or not some formal raw coordinates coincide. At stage four use the preceding anisotropic cover, valid throughout $[0, 1/2]$ because its ten exponents are distinct at every positive calibration in that interval. At stage five, $m = 2$ and the full-future cover has squared radius at most $C \max\{M^{-1/2}, \theta^{2/5} M^{-2/5}\} \leq CM^{-1/3}$. Stage six has two coordinates and squared covering radius at most CM^{-1} . Formula (62) remains valid with all formal labels at every stage, even when values coincide. Its uniform Lipschitz estimate and the repeated-quantization argument prove the upper bound in (64) on the entire interval. The stage-four checkpoint lower bound proves the other direction.

Equating the two terms in (64) gives $M = \theta^{-6}$ and common order θ^2 . These are comparison orders, not a numerical integer threshold. At fixed positive calibration the rate is $M^{-1/4}$, consistently with eight attainable dimensions; the ambient nine-dimensional rate $M^{-2/9}$ does not occur. $\square$

APPENDIX E. A UNIFORM STREAMING LAW THROUGH AN ADDITIVE COLLISION

We apply the confluent law to a family in which exact dimension changes at a fixed horizon and at a fixed positivity margin. In this family all repeated quantization errors can be controlled in the physical prediction metric, uniformly at the collision.

E.1. One fixed five-trial protocol. Let $0 \leq \theta \leq 1/2$, $A_\theta = \{0, 1, 2 + \theta\}$, and let μ be any fixed full-support probability on $[0, 1]$. Define

$$(65) \quad \begin{aligned} k_1^\theta &= \frac{1}{3} + \frac{1}{12}t, & k_2^\theta &= \frac{1}{3} + \frac{1}{12}t^{2+\theta}, \\ k_0^\theta &= \frac{1}{3} - \frac{1}{12}(t + t^{2+\theta}). \end{aligned}$$

These cells are at least $1/6$, sum to one, and span W_{A_θ} . Use the fixed comparator cube $[1/4, 3/4]^3$; every report factor is at least $\kappa = 1/24$. The three fixed probe commands have failure factors

$$(66) \quad F_j^\theta = \frac{1}{2} + \frac{1}{8}k_j^\theta, \quad j = 0, 1, 2.$$

The command accepts cell j with probability $3/8$ and the other cells with probability $1/2$. Their sum is $13/8$, so these factors span W_{A_θ} . At a checkpoint n use the 3^{5-n} ordered product queries of length $5 - n$. Exactly one checkpoint and one query are used in a run; all five trials are counted.

For the unconditional experiment the first three commands are independent uniform vectors in the comparator cube. After their reports have been compressed, reveal the independent uniform two-step query. The controller knows θ and μ as read-only calibration, but has access to past commands and reports only through its persistent index. The input law and all query labels are independent of θ .

For a history write $M_s = \mu(P_h t^s)$ and use the five coordinates

$$(67) \quad w = (M_1, M_2, M_{3+\theta}, M_{4+2\theta}, M_{2+\theta} - M_2).$$

They determine every length-two future law. They are ordinary posterior expectations of physical tests and their differences. For analysis put

$$(68) \quad z = (w_1, w_2, w_3, w_4, w_5/\theta) \quad (\theta > 0), \quad z_5|_{\theta=0} = \mu(P_h t^2 \log t).$$

This fifth coordinate at zero is an analytic continuation, not an extra observable supplied to the filter. The zero-calibration future law uses only the first four coordinates.

Theorem E.1 (Sharp two-parameter streaming order). *Set*

$$(69) \quad \Psi_\theta(M) = \max\{M^{-1/2}, \theta^{2/5} M^{-2/5}\}, \quad M \geq 1,$$

where the second term is zero at $\theta = 0$. There are constants $0 < c \leq C < \infty$, depending on the fixed prior and protocol but not on M or $\theta \in [0, 1/2]$, with the following properties.

There is a deterministic M -state filter, updated after every report, whose conditional prediction regret at every checkpoint and every admitted history is at most $C\Psi_\theta(M)$. In the prescribed three-trial exploration experiment, every M -state filter has unconditional regret at least $c\Psi_\theta(M)$. The same lower bound therefore holds for the minimax criterion over histories and checkpoints. The lower assertion permits the independent randomization of Definition 3.1.

In particular the two regimes are

$$(70) \quad \overline{\mathcal{R}}_{M,5}^{\theta,\text{stream}} \asymp \begin{cases} M^{-1/2}, & 1 \leq M \leq \theta^{-4}, \\ \theta^{2/5} M^{-2/5}, & M \geq \theta^{-4}, \end{cases}$$

with uniform comparison constants. This is an order-sharp crossover, not a claim of an exact transition budget or a leading distortion constant. The family and its base-model perturbation estimate were introduced in the source-pinned technical note [14]. The conclusion here is the matching, uniform two-parameter law and its online implementation.

E.2. Attainment of the fifth desingularized direction.

Lemma E.2 (A nondegenerate limiting patch). *At a common interior three-command tuple the map to z in (68) has rank five for every $\theta \in [0, 1/2]$. Its all-failure subprobability law dominates a fixed positive multiple of uniform probability on a translated five-dimensional cube of fixed side length, uniformly in θ .*

Proof. The formal two-fold sum pairs give exponents

$$0, 1, 2, 2 + \theta, 3 + \theta, 4 + 2\theta.$$

There is just one double cluster, at exponent two. Its divided difference is $(t^{2+\theta} - t^2)/\theta$. The nonconstant orders are $(0, 0, 0, 0, 1)$. The condition in (48) is $5 \leq 3(3 - 1) = 6$.

For an explicit common attainable tuple take $c_1 = 1/64$, $c_2 = 2/64$, $c_3 = 3/64$ and failure factors $(1 + c_i t^{2+\theta})/2$. They are realized by rejection probabilities

$$(1 - g_0, 1 - g_1, 1 - g_2) = (1/2 - 2c_i, 1/2 - 2c_i, 1/2 + 4c_i),$$

strictly inside $[1/4, 3/4]^3$. At zero their product tangent is the entire polynomial space of degree at most six. The future confluent space has basis

$$1, \quad t, \quad t^2, \quad t^2 \log t, \quad t^3, \quad t^4.$$

Lemma 5.2 makes its mixed pairing with that tangent surjective. Normalization removes one dimension, not two. For example, for the uniform prior, the minor with rows t^i , $0 \leq i \leq 5$, is exactly

$$(71) \quad \det \left[\frac{1}{i+1}, \frac{1}{i+2}, \frac{1}{i+3}, -\frac{1}{(i+3)^2}, \frac{1}{i+4}, \frac{1}{i+5} \right]_{i=0}^5 = \frac{1}{338751673344000000} > 0.$$

This rational identity is only a diagnostic of the preceding all-prior argument.

For every positive $\theta \leq 1/2$ the six displayed exponents are distinct and the binomial tangent has seven distinct exponents. Strict mixed-moment positivity gives surjectivity throughout this interval. All derivatives in the desingularized coordinates extend continuously to zero, as proved in Lemma B.2. Compactness and the uniform inverse construction in Lemma B.3 therefore apply on the *whole* interval $[0, 1/2]$, not only on an unspecified punctured neighborhood. The exact change of order in (68) is a fixed permutation of the general jet coordinates and changes no rank or minorization conclusion. $\square$

E.3. Uniform causal quantization.

Proof of Theorem E.1. At times one and two retain, exactly for the moment, the chronological normalized factors $f_i/\mu f_i$, with two coefficient coordinates per factor. At time three change to w in (67). At time four retain $(M_1, M_{2+\theta})$, and at time five use the unique terminal state. These states are sufficient for the remaining observable future law. All their reachable sets are compact.

The first two factor sets lie in fixed boxes of dimensions two and four, uniformly in θ . Indeed report coefficients are uniformly bounded, $\mu f \geq \kappa$, and the coefficient hyperplane $\mu f = 1$ can be coordinatized by the two nonconstant coefficients, its constant coefficient then being $1 - \sum_{a>0} f_a \mu(t^a)$. The required prior moments are fixed read-only data for each known calibration. Uniform boundedness of these affine formulas on $[0, 1/2]$ is immediate.

At time three the first four coordinates of w belong to $[0, 1]$. For $0 < t \leq 1$, the fundamental theorem of calculus in the exponent gives

$$|t^{2+\theta} - t^2| \leq \theta t^2 |\log t| \leq \theta.$$

This extends to zero. Consequently $-\theta \leq w_5 \leq 0$ for every posterior, with no evidence denominator in this bound. At time four both retained moments lie in $[0, 1]$. Lemma B.4, with reachable representatives in nonempty cells, now gives at most M representatives at each time and squared covering errors

$$(72) \quad \delta_1^2 \leq C_1 M^{-1}, \quad \delta_2^2 \leq C_2 M^{-1/2}, \quad \delta_3^2 \leq C_3 \Psi_\theta(M), \quad \delta_4^2 \leq C_4 M^{-1}.$$

The C_i are independent of θ . At zero simply omit the last, constant coordinate of the third state. Every bound in (72) is at most $C_0 \Psi_\theta(M)$.

It remains to verify that repeated updates do not divide by θ . The factor-to-factor update appends the current normalized factor. The factor-to- w update uses the product of the two retained factors and the new report factor. It is a ratio of polynomial coefficient expressions and bounded prior integrals. Along line segments between factor tuples every interpolated normalized factor is at least κ , so the evidence denominators and derivatives have positive uniform bounds. Each physical test in w is uniformly bounded; differentiating in factor coordinates therefore gives a uniform Lipschitz constant for this changeover.

For the only nontrivial shrinking update, write the new report factor as $f = a + bt + ct^{2+\theta}$. In terms of w set

$$Z_f = a + bw_1 + c(w_2 + w_5).$$

The next two moments are exactly

$$(73) \quad M'_1 = \frac{aw_1 + bw_2 + cw_3}{Z_f}, \quad M'_{2+\theta} = \frac{a(w_2 + w_5) + bw_3 + cw_4}{Z_f}.$$

Coefficients are uniformly bounded and $Z_f \geq \kappa$. A line segment between reachable w vectors represents a posterior mixture, so the same lower bound holds along that segment. The quotient rule proves a uniform Lipschitz estimate in the *physical* w coordinates. There is no divided difference and no factor $1/\theta$ in (73). The next, terminal update is constant.

At each time the filter stores only a representative index and applies the exact transition to that representative and the current input, then requantizes. A reachable representative followed by any admitted report has a reachable image, so every quantizer is used inside its domain. If e_n is the state error in the appropriate Euclidean physical coordinates, the preceding bounds give

$$e_0 = 0, \quad e_{n+1} \leq L_n e_n + C_0^{1/2} \Psi_\theta(M)^{1/2},$$

with L_n independent of θ . Iterate this recurrence through the five times. At times one and two each query prediction is a normalized polynomial expression in retained factors; at times three and four it is a fixed linear expression in the retained physical moments. These prediction maps have uniform Lipschitz constants, again by the positive evidence bounds. Squaring the resulting prediction error proves the conditional upper bound at every checkpoint. The representatives and transition functions are supplied existentially from the exact model, not synthesized from finite-precision exponent data. Only their index is persistent mutable memory.

For the lower bound at time three apply Lemma E.2 and the physical-metric argument in the proof of Theorem B.1, now to the nine common queries in (66). Their coefficient change from any other fixed spanning basis is invertible before taking products, so its full product expansion has constant full column rank. Four physical directions have scale one and the last has scale θ . Lemma B.4 gives the maximum of the four-dimensional lower bound $cM^{-1/2}$ and the five-dimensional lower bound $c\theta^{2/5}M^{-2/5}$. The constants are uniform on $[0, 1/2]$. Every streaming filter is an M -message encoder at this time, and the retained all-failure evidence is at least $(1/4)^3$. The randomization argument of Theorem B.1 applies unchanged. Finally, minimax regret is at least the unconditional regret in this one prescribed experiment. This completes the proof. $\square$

Corollary E.3 (Uniform bit law and the lower-constant scale). *For sufficiently small $\varepsilon > 0$, the least persistent bits for unconditional regret at most ε in the five-trial protocol, and likewise for its checkpoint-uniform minimax criterion, satisfy*

$$(74) \quad B_*(\varepsilon, \theta) = \log_2 \max\{\varepsilon^{-2}, \theta\varepsilon^{-5/2}\} + O(1),$$

where the $O(1)$ is uniform in $\theta \in [0, 1/2]$. If

$$c_{\text{opt}}(\theta) = \inf_{M \in \mathbb{N}, M \geq 1} M^{2/5} \overline{\mathcal{R}}_{M,5}^{\theta, \text{stream}},$$

then $c_{\text{opt}}(\theta) \asymp \theta^{2/5}$ for $0 < \theta \leq 1/2$, with constants independent of θ .

Proof. Solve the two necessary and sufficient inequalities from Theorem E.1 for M , obtaining, up to constant factors, $M \geq \varepsilon^{-2}$ and $M \geq \theta\varepsilon^{-5/2}$. Rounding to $M = 2^B$ changes B by a bounded amount. The same bounds apply to the minimax criterion. The lower bound for c_{opt} follows from the five-dimensional term. For its upper bound let M tend to infinity in the

matching upper estimate, so $M^{-1/10}$ disappears. No limiting leading quantization constant is needed. $\square$

The exact profiles remain $(0, 2, 4, 4, 2, 0)$ at zero and $(0, 2, 4, 5, 2, 0)$ at positive θ . The new assertion is that the extra direction is resolved at distortion scale θ^2 and state-budget scale θ^{-4} , with matching estimates on both sides. It sharpens the base-model bound $C(M^{-1/2} + \theta^2)$ without contradicting it.

APPENDIX F. THE VALUE OF A RESOLVED DIRECTION IN ONE SEQUENTIAL DECISION

We next use the same exploration histories, the same two-step query menu and the same eventual Bernoulli event as in Section E. Only the terminal decision is changed. This gives a common-payoff comparison between keeping and erasing the newly resolved direction, rather than comparing two separately normalized prediction tasks.

F.1. An independently priced observable ticket. After the time-three state has been formed, reveal the query J and a price U , independent and uniform on $[0, 1]$. The controller chooses a binary action $A \in \{0, 1\}$. Execute the two query trials and let Y be the same all-failure indicator as before. The payoff is

$$(75) \quad A(Y - U).$$

The price and the query are new inputs, not historical storage. The payoff belongs to $[-1, 1]$ and depends only on observable apparatus outcomes and the chosen action. Accepting the ticket changes the payoff, not the detector law or the five-trial budget. The exploration command program is prescribed; it is not optimized by the controller.

Let $p_j(h) = \mathbb{P}(Y = 1 \mid h, J = j)$. Write V_*^θ for the full-history optimal expected payoff and V_M^θ for the supremum over all M -state streaming policies, including randomized transitions and terminal actions. At the terminal decision the policy reads only its index, the new query and the new price.

Proposition F.1 (Exact decision–prediction identity). *For every fixed streaming encoder with index I , its optimal terminal rule accepts exactly when $U \leq \mathbb{E}[p_J(h) \mid I, J]$. Its value gap is*

$$(76) \quad V_*^\theta - V^\theta(I) = \frac{1}{2s} \sum_{j=1}^s \mathbb{E} \left[(p_j(h) - \mathbb{E}[p_j(h) \mid I])^2 \right], \quad s = 9.$$

Consequently

$$(77) \quad c\Psi_\theta(M) \leq V_*^\theta - V_M^\theta \leq C\Psi_\theta(M)$$

with positive constants independent of M and $\theta \in [0, 1/2]$.

Proof. Conditional on the full history and $J = j, U = u$, action one has expected payoff $p_j - u$ and action zero has payoff zero. The optimum is the threshold rule and its average over price is $\int_0^{p_j} (p_j - u) du = p_j^2/2$. Given only an index, replace p_j by $\bar{p}_{I,j} = \mathbb{E}[p_j \mid I]$, since price and query are independent of the prefix and encoder. A randomized action is a convex combination of the two payoffs and cannot improve the threshold optimum. Thus

$$V_*^\theta = \frac{1}{2s} \sum_j \mathbb{E} p_j^2, \quad V^\theta(I) = \frac{1}{2s} \sum_j \mathbb{E} \bar{p}_{I,j}^2.$$

Conditional-variance decomposition gives (76). Conditional expectation is also the optimal squared-loss decoder for the same encoder. Taking the infimum over the same streaming encoders therefore makes the optimal value gap exactly one half the optimal prediction regret. Theorem E.1 proves (77). The upper implementation may also threshold the representative's

prediction directly: conditional on a history, integrating the mistaken decisions over the interval between that prediction and p_j gives half their squared difference. $\square$

This proposition is a decision identity, not a new general comparison principle. Its use here is to retain the entire resolved streaming law in a bounded, binary-action task and then compare a specific erased observation in that *same* task.

F.2. Erasing the weak coordinate. Let

$$(78) \quad T_\theta(h) = (M_1, M_2, M_{3+\theta}, M_{4+2\theta})$$

be the first four coordinates of (67). This is a deterministic, calibration-dependent postprocessing of the history. It is not a coarsening of a raw one-step detector symbol, and it is not claimed to be the best possible four-dimensional encoding. Let V_T^θ be the optimal payoff of a terminal policy given the *entire exact* statistic T_θ , with no memory restriction, and then the same J, U . This comparison gives the erased controller more computational and storage power than any M -message compression of T_θ .

Theorem F.2 (Uniform value of the weak direction). *There are constants $c_e, C_e, K > 0$, depending on the fixed prior and protocol but independent of $0 < \theta \leq 1/2$, such that*

$$(79) \quad c_e \theta^2 \leq V_*^\theta - V_T^\theta \leq C_e \theta^2.$$

Moreover, whenever $M \geq K\theta^{-4}$,

$$(80) \quad V_M^\theta - V_T^\theta \geq \frac{1}{2} c_e \theta^2 > 0.$$

In particular an M -state streaming policy beats every policy whose only historical input is an M -message compression of T_θ , even though V_T^θ allows that comparator uncompressed access to the four coordinates. Both values concern the same actual calibration, exploration law, query, price, target, payoff and five-trial budget. At $\theta = 0$ the statistic T_0 is sufficient and $V_^0 = V_T^0$.*

Proof. Put $Z_\theta = w_5/\theta$ for positive θ , with its continuous value at zero as in (68). The common product query expansion has the form

$$(81) \quad p_j(h) = a_j + \ell_j(T_\theta(h)) + \theta b_j Z_\theta(h),$$

where a_j, b_j and the linear forms ℓ_j are independent of θ . They are obtained by expanding the two factors in (66) into the six formal monomials and replacing $M_{2+\theta}$ by $M_2 + \theta Z_\theta$. The full coefficient matrix has rank six, as proved in Lemma B.2. Hence $\sum_j b_j^2 > 0$; otherwise its weak column would be zero.

Apply the conditional-variance identity from Proposition F.1 to the sigma-algebra generated by T_θ . With $b_* = (2s)^{-1} \sum_j b_j^2 > 0$ one obtains the exact formula

$$(82) \quad V_*^\theta - V_T^\theta = b_* \theta^2 \mathbb{E} \text{Var}(Z_\theta \mid T_\theta).$$

The random variable Z_θ is bounded uniformly, since $|(t^{2+\theta} - t^2)/\theta| \leq t^2 |\log t| \leq 1$ before taking any posterior expectation. Thus the conditional variance is uniformly bounded above.

For the lower bound, Lemma E.2 gives a fixed positive subprobability multiple β of uniform measure on a translated cube of side 2ρ in the coordinates (T_θ, Z_θ) . For every Borel function v of the first four coordinates,

$$\mathbb{E}|Z_\theta - v(T_\theta)|^2 \geq \beta \int |z - v(t)|^2 d \text{Unif}_{\text{cube}}(t, z) \geq \beta \rho^2 / 3.$$

The last inequality integrates the variance of the independent uniform last coordinate; it is unaffected by the translation. Taking the infimum over v proves $\mathbb{E} \text{Var}(Z_\theta \mid T_\theta) \geq \beta \rho^2 / 3$. This

argument uses domination of an actual distribution, not an assumed uniform posterior on a formal tangent space. Equation (82) proves (79) with uniform positive constants.

By (77),

$$V_M^\theta - V_T^\theta \geq c_e \theta^2 - C \max\{M^{-1/2}, \theta^{2/5} M^{-2/5}\}.$$

For $M \geq K\theta^{-4}$ the maximum on the right is at most $\theta^2 \max\{K^{-1/2}, K^{-2/5}\}$. Choose $K \geq 1$ large enough that $C \max\{K^{-1/2}, K^{-2/5}\} \leq c_e/2$. This proves (80). A policy observing only a compression of T_θ cannot outperform the one observing all of T_θ : conditional on that statistic its randomization is independent of the latent history, and averaging produces an admissible randomized T_θ -policy. Finally at zero the weak column in (81) vanishes, proving sufficiency and equality of the full and erased values. $\square$

The absolute advantage vanishes quadratically as the exponents collide. What remains nondegenerate is the advantage after normalization by the actual weak-direction scale θ^2 , once the state budget resolves that scale. The theorem is not a lower bound for arbitrary exploration objectives. It is an optimal-policy comparison in one prescribed sequential acquisition experiment, with no change of Bayes baseline and no substitution of the separate mechanical sign experiment.

Corollary F.3 (Necessary and sufficient budget order). *Define*

$$M_{\text{beat}}(\theta) = \min\{M \in \mathbb{N} : V_M^\theta > V_T^\theta\}.$$

For $0 < \theta \leq 1/2$ this minimum exists and $M_{\text{beat}}(\theta) \asymp \theta^{-4}$, with uniform constants. More explicitly, for some $0 < k < K < \infty$,

$$\begin{aligned} 1 \leq M \leq k\theta^{-4} &\implies V_M^\theta - V_T^\theta \leq -C_e \theta^2 < 0, \\ M \geq K\theta^{-4} &\implies V_M^\theta - V_T^\theta \geq \frac{1}{2}c_e \theta^2 > 0. \end{aligned}$$

The first assertion is vacuous if its integer range is empty.

Proof. Write $E_\theta = V_*^\theta - V_T^\theta$ and $L_M^\theta = V_*^\theta - V_M^\theta$. Theorems already proved give $E_\theta \leq C_e \theta^2$ and $L_M^\theta \geq c_s M^{-1/2}$. Their exact difference uses the same experiment and baseline, hence

$$V_M^\theta - V_T^\theta = E_\theta - L_M^\theta \leq C_e \theta^2 - c_s M^{-1/2}.$$

Choose $k = (c_s/(2C_e))^2$, decreasing it if necessary so that $k < K$. The first inequality follows. The second is Theorem F.2, with K increased if needed. The admissible filter classes are nested as M increases because extra states can be unused. Thus the first-crossing budget exists; the two bounds and integer rounding give its stated order. $\square$

This consequence was extracted in the source-pinned v7 referee technical note [19]. It concerns beating the exact, uncompressed T_θ comparator. It does not locate an exact integer transition, and it does not assert equality of the full and erased experiments at smaller *equal compressed* budgets. At zero T_0 is sufficient, so no finite-state policy has value exceeding $V_T^0 = V_*^0$.

APPENDIX G. A COLLISION BIT AT A COMMON FINITE-MEMORY BUDGET

This section compares total risks for one fixed target, not regrets against two different Bayes forecasters. It concerns an uncensored raw detector, so that erasure is literal postprocessing. It does not identify this operation with erasure *before* a commanded comparator.

G.1. The counted mechanical realization. Let the triangular lattice have fundamental area $\mathfrak{a} = \sqrt{3}/2$, let $I_R = [9/20, 47/100]$, and put a reflecting disk of radius R at each lattice point. In each cartridge prepare a uniform position in the ambient torus, a uniform velocity direction, and independent uniform apparatus coordinates. A guarded insertion fails when the position is inside a disk. Otherwise a unit-speed particle is released. Every attempted insertion consumes one cartridge.

Fix $0 < T < 3/50$ and $\epsilon > 0$ with $\epsilon(47/100)^2 < 1$. Distinct lifted disks have separation at least $3/50$; an outgoing reflected ray cannot reenter the same convex disk before another collision. Thus each released particle has at most one collision by time T . At a collision let z be its impact coordinate relative to the disk center. The fixed detector reports $+$ when its prepared uniform coordinate is at most $1/2 + \epsilon|z|^2/\pi$, and otherwise reports $-$. The controller never receives z or the uniform seed. This is an internally exact-coordinate apparatus, not a lower bound on the complexity of physical measurement.

For completeness, incoming one-collision states are parameterized by

$$q = Rn_\alpha - sv^-, \quad v^- = -\cos\phi n_\alpha + \sin\phi t_\alpha, \quad 0 < s < T, \quad |\phi| < \pi/2.$$

The backward segment remains outside its own disk by convexity and cannot reach a distinct disk within T by separation. The forward reflected segment has the same exclusion. Every regular colliding initial state has its unique impact triple. At fixed incoming laboratory angle the position Jacobian is $R \cos\phi$, and the angular coordinate change has determinant of absolute value one. Integration against ambient preparation $dq d\vartheta/(2\pi\mathfrak{a})$ gives collision mass

$$\frac{R}{2\pi\mathfrak{a}} \int_0^T \int_0^{2\pi} \int_{-\pi/2}^{\pi/2} \cos\phi d\phi d\alpha ds = \frac{2TR}{\mathfrak{a}}.$$

Tangencies are null, not an excluded positive-mass neighborhood. Since $|z| = R$ at impact, the four uncensored raw probabilities are

$$(83) \quad \begin{aligned} k_\partial &= \pi R^2/\mathfrak{a}, & k_0 &= 1 - \pi R^2/\mathfrak{a} - 2TR/\mathfrak{a}, \\ k_\pm &= TR/\mathfrak{a} \pm 2T\epsilon R^3/(\pi\mathfrak{a}). \end{aligned}$$

They are positive on I_R and sum to one. For k_0 , positivity follows already at $R = 47/100$, $T = 3/50$, using $\mathfrak{a} > 0.86$ and $\pi < 22/7$; the other inequalities follow from the stated bounds. Their coefficient matrix in degrees zero through three is

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & -2T/\mathfrak{a} & T/\mathfrak{a} & T/\mathfrak{a} \\ \pi/\mathfrak{a} & -\pi/\mathfrak{a} & 0 & 0 \\ 0 & 0 & 2T\epsilon/(\pi\mathfrak{a}) & -2T\epsilon/(\pi\mathfrak{a}) \end{pmatrix},$$

with determinant of absolute value $4T^2\epsilon/\mathfrak{a}^3$. A bounded lookup comparator therefore realizes $A = \{0, 1, 2, 3\}$. Merging the signs gives the three probabilities $k_\partial, k_0, k_+ + k_-$, which span $A = \{0, 1, 2\}$. This supplies the dense special cases of Theorem 1.3 from one counted apparatus. The stronger complete-record response, positive filter, boundary-control and surrogate derivations are preserved in the source companion [9]; the collision calculation above uses their whole-preparation normalization rather than conditioning away insertion failures.

G.2. One target and all state budgets. Consume two cartridges without censoring. Let $X \in \{\partial, 0, +, -\}$ be the first raw report, and let Y be the placement-failure indicator on the second cartridge. After the first report the controller stores one of M states, then announces its forecast $a \in [0, 1]$ and incurs $(a - Y)^2$. The erased experiment observes $e(X) \in \{\partial, 0, H\}$ with $e(+)=e(-)=H$. Both experiments use the same prior, both ambient attempts, and

the same second-cartridge variable Y . There is no acceptance charge or saved cartridge. This is a separate uncensored experiment; it is not subject to the lower acceptance margin of the streaming theorem.

Put $h(R) = \pi R^2/\mathfrak{a}$ and define

$$p_x = \mu k_x > 0, \quad z_x = \mu(k_x h), \quad b_x = z_x/p_x, \quad p_B = \sum_{x \in B} p_x, \quad z_B = \sum_{x \in B} z_x.$$

For a nonempty subset B set

$$(84) \quad \delta(B) = \sum_{x \in B} \frac{z_x^2}{p_x} - \frac{z_B^2}{p_B} = \frac{1}{p_B} \sum_{\{x,y\} \subset B} p_x p_y (b_x - b_y)^2.$$

All these quantities are finite combinations of prior moments through order five and the exact physical constants.

Theorem G.1 (Complete common-risk finite-state comparison). *For either alphabet $\mathcal{X}$ (four raw symbols or three erased tags), its optimal M -state risk, allowing independent randomization, is*

$$(85) \quad R_M(\mathcal{X}) = \mu h - \max_{\substack{\mathcal{Q} \text{ partition of } \mathcal{X} \\ |\mathcal{Q}| \leq M}} \sum_{B \in \mathcal{Q}} \frac{z_B^2}{p_B}.$$

A maximizing partition, its block index and block forecasts z_B/p_B give an attained finite-state implementation.

Let R_M^+ and R_M^0 denote the raw and erased risks and $\Gamma_M = R_M^0 - R_M^+$. Then $\Gamma_M \geq 0$ for every M , $\Gamma_1 = 0$, and, with $\Delta = \delta(\{+, -\})$,

$$(86) \quad \Gamma_3 = \Delta - \min_{x \neq y} \delta(\{x, y\}), \quad \Gamma_M = \Delta \quad (M \geq 4).$$

The case $M = 2$ is exactly (85): it compares the seven two-block raw partitions and the three two-block erased partitions (one-block partitions are dominated). Equivalently, if D_M^+ and D_M^0 are the minimum sums of merge costs in the two alphabets, then

$$(87) \quad \Gamma_M = \Delta + D_M^0 - D_M^+.$$

These are common total Bayes risks, not two separately normalized memory regrets.

Proof. Conditional on a block message, the optimal forecast is z_B/p_B . Expanding squared loss gives block contribution $z_B - z_B^2/p_B$; summing proves the risk of every deterministic partition. For arbitrary random encoders and decoders, first replace a randomized forecast at each message by its conditional mean. Holding these at most M forecasts fixed, each raw symbol can be assigned deterministically to a forecast minimizing its conditional squared loss. This cannot increase risk. Reoptimizing each resulting block forecast also cannot increase risk. Thus a deterministic partition is optimal, and the finite maximum is attained. A fresh public seed independent of X, Y merely averages such risks and does not improve the optimum.

The identity (84) is obtained by expanding the weighted variance. The lossless raw risk is $\mu h - \sum_x z_x^2/p_x$; the lossless erased risk exceeds it by Δ . Adding the optimal merging distortion in each alphabet gives (87). Any erased code is a raw code after the deterministic map e , proving $\Gamma_M \geq 0$ directly. With one state both forecasts equal μh . With at least four states both alphabets are losslessly encoded. With three states the erased alphabet is lossless, and an optimal raw partition can be refined to exactly three blocks without increasing risk. Such a partition merges exactly one pair. This proves (86). The partition counts for $M = 2$ are the numbers of nontrivial subsets modulo complementation, namely $(2^4 - 2)/2 = 7$ and $(2^3 - 2)/2 = 3$. $\square$

Write $\mu_j = \mu(R^j)$, let $\mu_H(dR) = R\mu(dR)/\mu_1$, and put

$$v = \text{Var}_{\mu_H}(R^2), \quad \bar{c} = 1/2 + (\epsilon/\pi)\mu_H(R^2).$$

The exact lossless gain retained from [9] is

$$(88) \quad \Delta = \frac{2T\mu_1}{\mathfrak{a}} \frac{(\epsilon v/\mathfrak{a})^2}{\bar{c}(1-\bar{c})} > 0.$$

Indeed conditional on a first hit the sign probability is $c(R) = 1/2 + \epsilon R^2/\pi$, and $\text{Cov}_{\mu_H}(h, c) = \epsilon v/\mathfrak{a}$. The two conditional-mean shifts are respectively this covariance divided by $\bar{c}$ and its negative divided by $1 - \bar{c}$. Their weighted squared sum is the covariance squared over $\bar{c}(1 - \bar{c})$. Multiplication by the hit probability proves (88), which is also precisely the pair cost $\delta(\{+, -\})$. Full support makes $v > 0$. This is an explicit conditional-variance calculation, not a new general comparison-of-experiments principle [1].

Corollary G.2 (A genuine finite-resolution threshold). *Fix $T > 0$ and a full-support prior on I_R . There exists $\epsilon_* > 0$ such that, for $0 < \epsilon < \epsilon_*$,*

$$(89) \quad \Gamma_M = 0 \quad (M = 1, 2, 3), \quad \Gamma_M = \Delta > 0 \quad (M \geq 4).$$

Thus a four-state, two-bit controller realizes the strict gain, whereas all smaller state budgets have exactly zero gain in this amplitude range. This does not identify exact algebraic dimension with useful accuracy.

Proof. At $\epsilon = 0$, $b_+ = b_- = b_H$. The three distinct tag means satisfy

$$b_0 < \mu h < b_H < b_\partial.$$

The first strict inequality follows because k_0 is strictly decreasing and h strictly increasing, so $\text{Cov}_\mu(k_0, h) < 0$. For example this sign follows by integrating $(k_0(R) - k_0(R'))(h(R) - h(R'))/2$ against two independent prior draws. The second inequality uses the positive covariance of R and h . The last follows in the hit-tilted prior from the same positive covariance of R and h , since the placement posterior is its additional R -tilt. Full support makes all these covariances strict.

For three states the unique zero-distortion merging pattern at amplitude zero is to keep the three distinct tag means separate and merge the two signs. Every other pair has positive merge cost. For two states, no optimal partition at amplitude zero splits the equal-mean signs between its two blocks. To prove this, use the optimal block means as centers. Equal-mean points assigned to different centers must be equidistant from them; otherwise reassignment to the closer center strictly improves risk. If the centers are distinct, b_H is their midpoint. Each block must then contain a different extreme tag to place its mean strictly on the corresponding side of b_H . Move one sign into the other block, holding centers fixed; the cost is unchanged. The vacated-sign block still contains its extreme tag, so reoptimizing its center strictly reduces the cost, a contradiction. If the two centers agree, the achieved risk is the one-state risk, strictly worse than separating one extreme tag. Thus an optimal two-state partition never splits the signs.

There are finitely many partitions. At amplitude zero every partition that splits the signs and uses at most two states has a strictly positive risk gap to the optimum, as does every non-sign-merging three-state partition. All their risks depend continuously on ϵ because the cell masses stay positive. Therefore, for a sufficiently small common positive ϵ_* , none of these partitions is optimal for $M = 2, 3$. Every remaining partition keeps the signs together and has exactly the risk of the corresponding erased partition, because both $p_+ + p_-$ and $z_+ + z_-$ are independent of ϵ . This proves equality of risks for $M = 2, 3$; $M = 1$ and $M \geq 4$ follow from Theorem G.1 and (88). $\square$

The threshold is checkable from finitely many moment expressions: enumerate the partitions in (85), find their positive gaps at zero, and bound their continuous rational derivatives while cell masses stay bounded away from zero. No prior-uniform or horizon-uniform threshold is claimed. In particular (88) can be extremely small for a concentrated prior. The threshold theorem gives an exact resource comparison, not a promise of a practically large gain.

APPENDIX H. CONVEX COLLISION ENERGIES AND PHASE BOUNDARIES

For completeness we give the finite allocation consequence of Theorem 7.6. This consequence, including nested optimizers and the ordering of phase boundaries, is recorded in the technical note [23]; it is not an additional independent contribution of the shared-memory theorem.

Proposition H.1 (Convexity of the collision energies). *Under the hypotheses of Theorem 7.6, set $\beta_0 = \beta_1 = 0$ and write $\beta_\ell = \beta_{m,\ell}$ up to the number q of distinct labels along the path. Then*

$$\beta_{\ell+1} - \beta_\ell \geq \beta_\ell - \beta_{\ell-1} \quad (1 \leq \ell < q).$$

Optimal subsets for these energies can be chosen nested in ℓ . For a past cap $p \leq q$ the adjacent asymptotic power boundaries of the checkpoint law, with M of order θ^{-b} , are

$$(90) \quad b_\ell = \ell\beta_{\ell+1} - (\ell+1)\beta_\ell, \quad 1 \leq \ell < p, \quad b_\ell \geq b_{\ell-1} \quad (2 \leq \ell < p).$$

Equal consecutive boundaries give an intermediate regime of zero length in the power parameter b .

Proof. The leaf cost sequence is $(0, 0)$ on its feasible range. At an internal cluster the recurrence (29) is

$$F_C(k) = \Delta_C \binom{k}{2} + \min_{\sum_D k_D = k} \sum_D F_D(k_D), \quad \Delta_C \geq 0.$$

Suppose inductively that each child sequence is discretely convex. Its successive marginal costs are then nondecreasing. The minimum convolution is obtained by taking the k cheapest members of the union of the child marginal lists. Ties can be resolved respecting predecessors within each list; the selection then corresponds to a feasible child allocation. Every other allocation selects k members and cannot have a smaller sum. Its successive marginal costs are therefore the sorted union of the child lists, a nondecreasing sequence. Adding $\Delta_C \binom{k}{2}$ adds $\Delta_C(k-1)$ to the k th marginal and preserves monotonicity. This additional term does not change the minimizing allocation at fixed k .

The prefix-respecting selection gives nested minimizing child counts. By the inductive nested-subset assertion in each child, their union can be chosen nested at C . Induction from the leaves to the root proves both assertions for $F_{\text{root}}(\ell) = \beta_\ell$.

For $M \asymp \theta^{-b}$ the ℓ th regret term has power $2(\beta_\ell + b)/\ell$ in θ . Equality of adjacent powers gives the first formula in (90). Subtraction gives

$$b_\ell - b_{\ell-1} = \ell(\beta_{\ell+1} - 2\beta_\ell + \beta_{\ell-1}) \geq 0.$$

The ordered adjacent intersections describe the lower envelope of these powers: below the next intersection the smaller adjacent index has the smaller power, and above it the larger index does. Iterating this comparison across the ordered boundaries proves that no nonadjacent term creates an intervening open regime. A repeated intersection thus has no interval on which the intermediate index alone is active. The past cap stops the list at p . These statements concern power orders; fixed comparison constants do not specify exact integer thresholds. $\square$

REFERENCES

- [1] D. Blackwell, *Equivalent comparisons of experiments*, Ann. Math. Statist. **24** (1953), 265–272. doi:10.1214/aoms/1177729032.
- [2] S. Eliahou and E. Mazumdar, *Iterated sumsets and Hilbert functions*, J. Algebra **593** (2022), 274–294. doi:10.1016/j.jalgebra.2021.11.019.
- [3] J. Elias, *Sumsets and projective curves*, Mediterr. J. Math. **19** (2022), article 177. doi:10.1007/s00009-022-02108-0.
- [4] S. Graf and H. Luschgy, *Foundations of Quantization for Probability Distributions*, Lecture Notes in Mathematics 1730, Springer, 2000. doi:10.1007/BFb0103945.
- [5] A. Kara and S. Yüksel, *Near optimality of finite memory feedback policies in partially observed Markov decision processes*, J. Mach. Learn. Res. **23** (2022), no. 11, 1–46. <https://jmlr.org/papers/v23/20-1152.html>.
- [6] M. L. Littman, R. S. Sutton and S. Singh, *Predictive representations of state*, Advances in Neural Information Processing Systems **14** (NIPS 2001). Original proceedings paper.
- [7] A. Pinkus, *Totally Positive Matrices*, Cambridge Tracts in Mathematics 181, Cambridge University Press, 2010.
- [8] J. Subramanian, A. Sinha, R. Seraj and A. Mahajan, *Approximate information state for approximate planning and reinforcement learning in partially observed systems*, J. Mach. Learn. Res. **23** (2022), no. 12, 1–83. <https://jmlr.org/papers/v23/20-1165.html>.
- [9] Q. Qi, *Finite-horizon information states of calibrated experiments*, English revision 5, September 6, 2026, source directory `papers/A1-english-v5` in TrillioniumFoundation/theta-theory, commit 20a26285a26c. Preserved verbatim in `papers/A1-english-v6/legacy/` in the repository.
- [10] *A1 English v5: referee report and claim audit*, owner-commissioned AI-assisted repository review, September 6, 2026, commit 171a7e20470d, `reviews/a1-english-v5-2026-09-06`. This is not a journal-appointed review or a proof certificate.
- [11] Q. Qi, *A1: strong finite-cylinder response on a common reporting space*, repository derivation, `round33/chapters/A1.tex`, commit 7f1bc9a42ba2.
- [12] C. de Boor, *Divided differences*, Surveys in Approximation Theory **1** (2005), 46–69. <https://arxiv.org/abs/math/0502036>.
- [13] Q. Qi, *Sparse observation algebras and finite-horizon memory*, English revision 6, September 6, 2026, commit 750a65ef6242.
- [14] *An exact dimension jump at fixed finite resolution*, technical note accompanying the owner-commissioned AI-assisted A1 v6 repository review, September 6, 2026, commit 414f8c43a623. This source introduced the near-collision family and its base-model bound; it did not assert the matching uniform crossover proved here.
- [15] D. Batenkov, B. Diederichs, G. Goldman and Y. Yomdin, *The spectral properties of Vandermonde matrices with clustered nodes*, arXiv:1909.01927v2 (2020), Theorems 2.2–2.3 and Corollary 2.1. <https://arxiv.org/abs/1909.01927>.
- [16] Y. Yomdin and G. Comte, *Tame Geometry with Application in Smooth Analysis*, Lecture Notes in Mathematics 1834, Springer, Berlin, 2004.
- [17] G. Comte and I. Halupczok, *Motivic Vitushkin invariants*, Compos. Math. **161** (2025), no. 5, 959–992. doi:10.1112/S0010437X25007031. The cited preprint is arXiv:2206.15412v2 (2024). The classical real variations and entropy inequality used here are recalled in the introduction, equations (4)–(5). <https://arxiv.org/abs/2206.15412>.
- [18] Y. Zhang and J. Kileel, *Covering Number of Real Algebraic Varieties and Beyond: Improved Bounds and Applications*, arXiv:2311.05116v4 (June 6, 2025), Lemma 2.18, p. 11. <https://arxiv.org/abs/2311.05116v4>.
- [19] *What the v7 theorems imply, and where their hypothesis matters*, technical note accompanying the owner-requested AI-assisted A1 v7 repository review, September 6, 2026, commit 72eb41e358bd. This note gives the necessary-budget deduction and the seven-trial past-limited boundary, not the attainable-flag classification proved here.
- [20] F. Leja, *Sur certaines suites liées aux ensembles plans et leur application à la représentation conforme*, Ann. Polon. Math. **4** (1957), no. 1, 8–13. <https://eudml.org/doc/208291>.
- [21] L. Bos, S. De Marchi, A. Sommariva and M. Vianello, *Computing multivariate Fekete and Leja points by numerical linear algebra*, SIAM J. Numer. Anal. **48** (2010), 1984–1999. doi:10.1137/090779024.
- [22] T. Guo, N. Karagodin and E. Stepanov, *Optimal functional product quantization*, author preprint, version `guokaragste22a8`, Sections 1.1–1.2 and 5.5. <https://cvgmt.sns.it/media/doc/paper/6002/guokaragste22a8.pdf>.
- [23] *The classical spectral content and phase structure of A1 v9*, source-pinned technical note accompanying the owner-requested AI-assisted repository review, September 6, 2026, commit 7a499e3cb323.